

Geometry-Driven Optimization of Passive UAV Relay Arrays for Beamforming and Null Synthesis

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ABSTRACT

This paper introduces and studies a mathematical modeling and computational optimization framework for passive UAV relay arrays used in directional beamforming and null synthesis. In contrast to active distributed beamforming architectures, the proposed framework does not require oscillator-level inter-element synchronization among active transmitters. Instead, for a known monochromatic incident field, the relative phases of the relay contributions are induced passively by propagation path differences determined by the array geometry. This leads naturally to a synthesis problem in which the three-dimensional placement of the passive relays is used to shape the far-field radiation pattern. We formulate this design task as a constrained nonlinear optimization problem over the admissible relay configurations, with objectives promoting energy concentration in prescribed beam directions and suppression in selected null directions, subject to geometric feasibility constraints such as minimum inter-relay separation. The associated objective functional depends continuously on the relay geometry, which yields an existence result for an optimal configuration by compactness of the admissible set. Thus, while the problem admits a global minimizer, the proposed algorithms are designed to find stable, high-quality local minima, which we benchmark against an extensive MultiStart search serving as an empirical near-global reference. Efficient numerical experiments further show that even relatively small passive UAV swarms can achieve substantial far-field pattern control through suitable spatial placement alone, highlighting the potential of passive mobile relay arrays for adaptive wireless signal shaping.

1. Introduction

Modern wireless systems are exploring the use of drone swarms as reconfigurable antenna arrays to overcome line-of-sight obstructions and dynamically direct signals. This idea has taken several forms, including swarms in which every drone carries its own array element [12]; single drones carrying reflecting elements or metasurfaces [30, 37]; active transmit and passive reflective realizations of such arrays [3, 8]; and combined architectures such as multi-hop aerial backhaul, in which a signal is relayed through several loci by a daisy chain [9], or through a broader network of satellites [15]. In each case, a transmitter radiates toward a drone or fleet of drones carrying such arrays, which redirect the energy toward an intended receiver. By appropriately manipulating each drone or array, the scattered signals add coherently in the desired direction and destructively interfere in others, producing a focused main lobe, deep nulls toward protected angles, and low sidelobes elsewhere [12].

The mechanism considered in this paper differs fundamentally from active distributed beamforming, and this distinction fixes the design variable for everything that follows. Here, the individual UAV elements do not act as synchronized transmitters; instead, for a known incident monochromatic field, each passive relay contributes a phase determined by its propagation path length and element response, and coherent addition in the far field is obtained purely through the geometric placement of the relays. The design variable throughout this paper is therefore the spatial configuration of the swarm, not an electronically controlled phase profile, and relay geometry is the primary tool for beam steering and null synthesis. The architecture is motivated by low-technology wireless scenarios in which a

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known or estimable incident field can be redirected or shaped without synchronized active transmitters. Covert low-probability-of-detection signaling, search-and-rescue communication support, and temporary field-deployable relay systems are examples where such an architecture may be useful.

Early work established how geometry and excitation shape beamwidth, sidelobes, and nulls. See Schelkunoff's array theory, Dolph–Chebyshev and Taylor distributions for low sidelobes, and the broader array processing canon [11, 19, 35, 40, 41]. The reflectarray and metasurface literature extends this to scalable, electronically tunable passive phase control [14]. Electronically steerable parasitic array radiator (ESPAR) antennas achieve a related effect at the element level, shaping radiation patterns with compact parasitic elements whose reactive loads are tuned near a driven source [27, 28]. Classical weight synthesis, reflectarrays, and ESPAR all begin with a favorable, uniform geometry and optimize the phase or reactance while keeping the spatial configuration fixed throughout.

Active synchronized transmission across the swarm, as opposed to passive redirection, comes with its own set of technical difficulties which are worth briefly mentioning. Translating classical beamforming to UAV swarms in this active mode imposes tight requirements on timing, frequency, phase alignment, and relative motion [26]. Tolerance analyses and correction strategies for micro-UAV arrays [31], together with early flight demonstrations of two- and three-drone distributed beamforming [10, 22], have validated feasibility at small scale, and recent advances in wireless synchronization — picosecond timing, internode ranging, and decentralized frequency alignment — push coherence further still [20, 21, 29, 33, 36]. Even so, robust coherence across many drones remains an open challenge [26]. Thus, using passive relay systems bypasses the stringent distributed coherence requirements of active arrays.

A more recent line of research treats the elements' positions as the design variables, a setting closer to our own. When positions rather than excitation weights are free, the map from geometry to far-field pattern is highly nonconvex; small spatial perturbations induce nonlinear phase changes creating an optimization landscape with many saddles and plateaus. Accordingly, much of this literature turns to global or stochastic search. Genetic algorithms, particle swarm optimization, and simulated annealing have been used to suppress sidelobes or place nulls [13, 34]. While useful, these stochastic methods are computationally expensive and lack guarantees of optimality or stable performance. Seeking a more rigorous formulation, Abdelhay [1] convexifies the problem for small element displacements via first-order approximations and sparse formulations, reducing peak sidelobes and enforcing nulls while moving as few elements as possible. The underlying geometry-to-pattern relationship, however, remains approximated rather than solved in its exact nonlinear form. Moving closer to the context of airborne arrays, Zarifi et al. [42] and Egarguin et al. [12] demonstrated that jointly optimizing the three-dimensional coordinates of reflector-equipped drones (along with element phases, when available) yields a strong main beam with low sidelobes and prescribed nulls. See also [18, 43, 44] for movable-antenna communication systems in which active antenna positions are jointly optimized alongside transmit or receive beamforming weights; however, of course, these mechanisms are active rather than passive. Closest in spirit to our approach is Liu et al. [17] which jointly optimizes the location and orientation of a single passive reflecting aperture. The present framework differs from all of these in that the aperture is not one surface on a single UAV nor a fixed layout with movable active elements, but a distributed set of passive dipole scatterers each carried (individually) by an element of the swarm.

This paper is organized as follows. We conclude this section with the main contributions of the paper. Section 2 derives the far-field equations relevant to our work. Section 3 formulates the optimization problem whose solution yields the optimal drone swarm arrangement for our beamforming objectives, and in its second half addresses the theoretical side of the problem: the design of smooth sidelobe suppression functionals, and a proof that a global minimizer to this highly nonconvex design problem exists. Section 4 introduces the algorithms used to solve the problem and our Ansatz approach for finding an initial configuration. Section 5 presents numerical results demonstrating our beamforming approach for a representative case ($N = 9$ drones). Section 6 discusses the implications of this work and directions for future research.

1.1. Contributions and Positioning

- We formulate the relay geometry design problem for a UAV swarm carrying passive scatterers directly as a constrained far-field pattern synthesis problem. This distinguishes the framework from active distributed arrays [24–26] and from aerial IRS/RIS designs [3, 7, 8, 32], where phase control is achieved electronically rather than geometrically.
- We prove that the constrained design problem admits a global minimizer, establishing well-posedness. The argument combines compactness of the feasible set with upper and lower hemicontinuity of the critical point

correspondence Γ (used to construct a measure of peak sidelobe level), an application of the Berge Maximum Theorem, and the implicit function theorem to handle non-degenerate critical points [2, 5].

- We introduce a concentric circular Ansatz that places elements on two offset rings with altitudes fixed by a beamfocusing condition. A coarse grid search over ring radii and offset angle identifies the best symmetric starting geometry. In `MultiStart` experiments with 10,000 random initializations, fewer than 0.72% achieved peak sidelobe levels below -25 dB, confirming that this structured initialization is critical to the success of local methods.
- We utilize (i) an exact trust-region solver to escape saddles and plateaus characteristic of nonconvex beampattern objectives and design (ii) a zoom-in penalty continuation scheme (ZIPC) which drives the more sensitive null depth objective to numerically negligible levels without undermining sidelobe suppression. Both methods are benchmarked against standard local and global solvers, including `fmincon`, `fminunc`, `ADAM`, and `MultiStart`. Across these experiments the proposed methods consistently return stable, high-quality local minima whose objective values are close to the best `MultiStart` results, so that `MultiStart` serves throughout as an empirical near-global benchmark.

2. Far-Field Derivation

We formalize the geometry to pattern mapping, inserting classical references where the steps are standard. We work in unbounded three-dimensional free space and assume a narrowband, time-harmonic regime with angular frequency ω and the $\exp(-i\omega t)$ sign convention throughout. Bold symbols (e.g., $\mathbf{r}$) denote vectors. We represent the n^{th} drone's position in Cartesian space as $\mathbf{r}_n = (x_n, y_n, z_n)$. Let $\mathbf{r}_s = (x_s, y_s, z_s)$ denote the center of the relay swarm in Cartesian coordinates and (r_s, θ_s, ϕ_s) for its spherical coordinates. We collect the positional coordinates as $\mathbf{x} := (x_1, y_1, z_1, \dots, x_N, y_N, z_N)^\top$ then define $\mathbf{r}'_n(\mathbf{x}) := \mathbf{r}_n(\mathbf{x}) - \mathbf{r}_s$ for $n = 1, \dots, N$ as the n^{th} drone's position relative to the swarm center. Let $\hat{\mathbf{r}}(\theta, \phi)$ be the unit vector in direction (θ, ϕ) , and define the plane wave vector

$$\mathbf{k}(\theta, \phi) = \frac{2\pi}{\lambda_0} (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta), \quad (1)$$

with λ_0 the free-space wavelength. We consider an incident plane wave arriving from a transmission source TX (generally a transmitter on the ground) in the direction (θ_s, ϕ_s) with wavevector $\mathbf{k}_{\text{in}} = \mathbf{k}(\theta_s, \phi_s)$ pointing towards the center of the swarm geometry. We observe the re-radiated field in direction (θ, ϕ) with $\mathbf{k}_{\text{out}} = \mathbf{k}(\theta, \phi)$ modeled as originating at the swarm center and being measured at a point RX (generally a receiver on the ground). The passive relay array geometry is illustrated in Figs. 1 and 2.

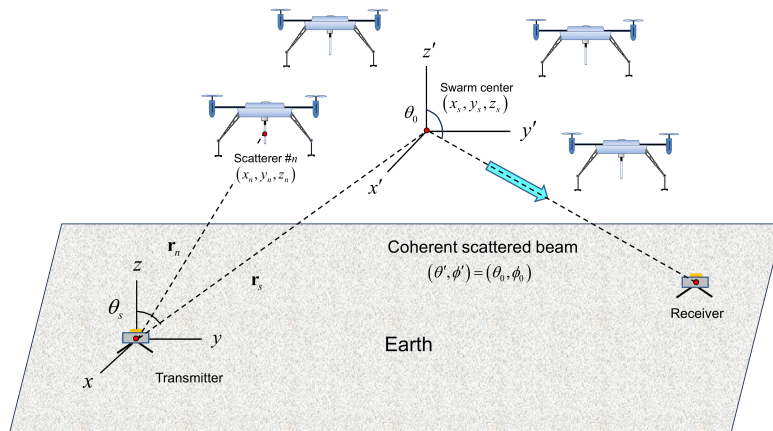


Fig. 1: Passive UAV relay-swarm geometry and coordinate conventions. The incident direction (θ_s, ϕ_s) is measured in the transmitter-centered unprimed coordinate system, whereas the outgoing direction (θ, ϕ) is measured in the primed coordinate system centered at the swarm. The desired coherent scattered-beam direction is (θ_0, ϕ_0) . The n^{th} UAV carries a z -directed dipole scatterer at $\mathbf{r}_n = \mathbf{r}_s + \mathbf{r}'_n$.

The scattered field is generated by the electric currents induced on the passive relay elements. From this point on, let $\mathbf{r} = R\hat{\mathbf{r}}$ and $\mathbf{r}'$ be coordinates measured from the swarm center, and let $\mathbf{J}(\mathbf{r}')$ denote the phasor of the induced volume current density. In the Lorenz gauge, the magnetic vector potential satisfies

$$(\nabla^2 + k^2)\mathbf{A} = -\mu_0\mathbf{J}, \text{ where } k = \frac{2\pi}{\lambda_0} \text{ denotes the wave number.}$$

For the $\exp(-i\omega t)$ convention, the outgoing scalar Green's function satisfies $(\nabla^2 + k^2)G = -\delta$ and is

$$G(\mathbf{r} - \mathbf{r}') = \frac{\exp(ik|\mathbf{r} - \mathbf{r}'|)}{4\pi|\mathbf{r} - \mathbf{r}'|}.$$

The Green's function acts componentwise on $\mathbf{J}$, and hence

$$\begin{aligned} \mathbf{A}(\mathbf{r}) &= \mu_0 \int_V G(\mathbf{r} - \mathbf{r}') \mathbf{J}(\mathbf{r}') d^3\mathbf{r}' \\ &= \frac{\mu_0}{4\pi} \int_V \frac{\exp(ik|\mathbf{r} - \mathbf{r}'|)}{|\mathbf{r} - \mathbf{r}'|} \mathbf{J}(\mathbf{r}') d^3\mathbf{r}'. \end{aligned} \quad (2)$$

The vector potential is given by Eq. (2). This is the magnetic vector potential produced by the induced current, rather than the electric field itself [16, 39].

In the Fraunhofer region, $|\mathbf{r} - \mathbf{r}'| \approx R - \hat{\mathbf{r}} \cdot \mathbf{r}'$. Moreover, because $\mathbf{E}^{\text{sc}} = i\omega\mathbf{A} - \nabla\Phi$, for the chosen time convention, (where Φ denotes the associated Lorenz-gauge scalar potential), the scalar-potential term cancels the longitudinal part of $\mathbf{A}$ to leading order. Define the transverse projector

$$\mathbf{P}_\perp(\hat{\mathbf{r}}) := \mathbf{I} - \hat{\mathbf{r}}\hat{\mathbf{r}}^\top, \quad \text{and,} \quad \mathbf{A}_\perp := \mathbf{P}_\perp(\hat{\mathbf{r}})\mathbf{A} = A_\theta\hat{\boldsymbol{\theta}} + A_\phi\hat{\boldsymbol{\phi}}.$$

The radiated electric field is therefore

$$\begin{aligned} \mathbf{E}^{\text{sc}}(\mathbf{r}) &= i\omega\mathbf{A}_\perp(\mathbf{r}) + O(R^{-2}) \\ &\approx \frac{i\omega\mu_0 \exp(ikR)}{4\pi R} \mathbf{P}_\perp(\hat{\mathbf{r}}) \tilde{\mathbf{J}}(\mathbf{k}_{\text{out}}), \\ \tilde{\mathbf{J}}(\mathbf{k}) &:= \int_V \exp(-i\mathbf{k} \cdot \mathbf{r}') \mathbf{J}(\mathbf{r}') d^3\mathbf{r}', \quad \mathbf{k}_{\text{out}} = k\hat{\mathbf{r}}. \end{aligned} \quad (3)$$

Thus, only the transverse (θ, ϕ) components of the magnetic vector potential contribute to the leading-order radiated electric field.

To recover the array factor, let $\mathbf{j}_e(\mathbf{u})$ denote the induced-current distribution on one reference element illuminated by the fixed incident plane wave, with the incident phase referenced to the center of that element. Under the plane-wave, identical-element, and negligible-mutual-coupling approximations, linearity and translation invariance give

$$\mathbf{J}(\mathbf{r}') = a_0 \sum_{n=1}^N \exp(i\mathbf{k}_{\text{in}} \cdot \mathbf{r}'_n(\mathbf{x})) \mathbf{j}_e(\mathbf{r}' - \mathbf{r}'_n(\mathbf{x})), \quad (4)$$

where a_0 is a common complex excitation coefficient containing the incident-field amplitude. If

$$\tilde{\mathbf{j}}_e(\mathbf{k}) := \int_{\mathbb{R}^3} \exp(-i\mathbf{k} \cdot \mathbf{u}) \mathbf{j}_e(\mathbf{u}) d^3\mathbf{u},$$

then the Fourier translation rule gives

$$\tilde{\mathbf{J}}(\mathbf{k}_{\text{out}}) = a_0 \tilde{\mathbf{j}}_e(\mathbf{k}_{\text{out}}) \sum_{n=1}^N \exp(-i[\mathbf{k}_{\text{out}} - \mathbf{k}_{\text{in}}] \cdot \mathbf{r}'_n(\mathbf{x})). \quad (5)$$

Define the vector element pattern, up to a direction-independent normalization, by

$$\mathbf{e}(\theta, \phi) := \mathbf{P}_\perp(\hat{\mathbf{r}}) \tilde{\mathbf{j}}_e(\mathbf{k}_{\text{out}}).$$

After suppressing the common factors a_0 , $i\omega\mu_0$, and $\exp(ikR)/(4\pi R)$, the far field becomes

$$\mathbf{E}_\infty(\theta, \phi; \mathbf{x}) \propto \mathbf{e}(\theta, \phi) \sum_{n=1}^N \exp(-i[\mathbf{k}_{\text{out}} - \mathbf{k}_{\text{in}}] \cdot \mathbf{r}'_n(\mathbf{x})). \quad (6)$$

This is the usual separation into an element pattern and an array factor [4, 14, 19]. Writing $e(\theta, \phi) := \|\mathbf{e}(\theta, \phi)\|_2$, the physical radiation intensity (power per unit solid angle) is proportional to the squared magnitude of this far field. We therefore use the following normalized radiation intensity in the optimization:

$$U(\mathbf{x}, (\theta, \phi)) = |e(\theta, \phi)|^2 \left| \sum_{n=1}^N \exp(-i[\mathbf{k}(\theta, \phi) - \mathbf{k}(\theta_s, \phi_s)] \cdot \mathbf{r}'_n(\mathbf{x})) \right|^2. \quad (7)$$

This is the quantity we shape by optimizing $\mathbf{x}$.

The derivation above holds for an arbitrary element pattern $e(\theta, \phi)$. For the numerical experiments in this paper we restrict to resonant z-directed half-wave dipole scattering elements (length $L = \lambda_0/2$) with a sinusoidal current distribution $I(z) = I_0 \cos(kz)$, $|z| \leq \lambda_0/4$. Balanis [4] derives the exact far-field amplitude pattern obtained by integrating this current distribution to be:

$$e(\theta) = \frac{\cos\left(\frac{\pi}{2} \cos \theta\right)}{\sin \theta}. \quad (8)$$

The scattered field remains linearly polarized in the $\hat{\theta}$ direction by the azimuthal symmetry of a z-directed source. The incident plane wave arriving from (θ_s, ϕ_s) is likewise taken to be vertically polarized, so that the z-projection that drives each dipole is $E_z^{\text{inc}} = E_0 \sin \theta_s$ where $E_0 = \sqrt{(2\pi)^{-1} \eta_0 P_t G_t(\theta_s, \phi_s) / R_t}$ is the amplitude of the θ -polarized incident field at the array center, with P_t the transmitted power, $G_t(\theta_s, \phi_s)$ the gain of the transmit antenna in the direction of the array, R_t the distance from the transmitter to the array center, and η_0 the free-space intrinsic impedance. The factor, E_z^{inc} , is absorbed into the overall amplitude (i.e., absorbed into a_0 in Eq. (4)) and therefore does not affect the array factor optimization. The pattern $e(\theta)$ vanishes at $\theta = 0$ and $\theta = \pi$ (along the dipole axis) and attains its maximum at the equatorial plane $\theta = \pi/2$. Although e has a removable singularity at the poles, the radiation intensity U depends on e^2 , which extends smoothly to zero at the poles. We therefore assume e^2 is smooth on the sphere, which holds for the half-wave dipole pattern used throughout.

The model described in Eqs. (3) to (6) captures the dominant geometry-induced phase effects responsible for far-field beamforming and null synthesis. Mutual coupling, platform scattering, finite UAV body effects, and orientation-dependent polarization mismatch are not fully resolved in this reduced model. The minimum inter-relay spacing constraint of $d_{\min} = \lambda_0/4$, which we will later introduce in Section 3.4, is therefore best interpreted as a practical mechanism for mitigating, rather than eliminating, mutual coupling. A full-wave coupled-dipole or Method-of-Moments validation of the optimized configurations is a natural next step, but it is outside the scope of the present study.

3. Optimization Problem Formulation

In this section we formulate the optimization problem which yields the desired pattern properties and provide proof of global minima existence.

3.1. Problem Geometry

Fig. 2 provides a 2D view of the full Passive Relay Array (PRA) system, emphasizing the distances (R_t , R_r), and elevation angles between the transmitter, the PRA, and the receiver relative to the earth. All system model variables definitions and their units can be found in Table 1.

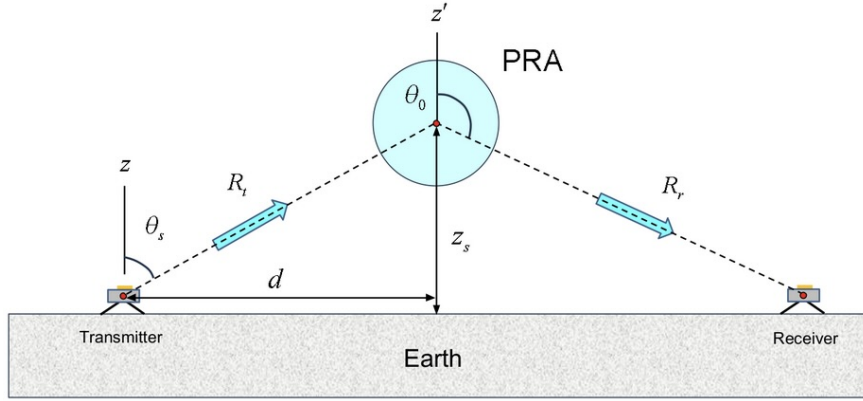


Fig. 2: Passive Relay Array System (2D View)

Table 1

Definition of variables used in the system model.

Symbol	Description	Units
θ_s	Elevation angle of the PRA relative to the TX	radians
ϕ_s	Azimuth angle of the PRA relative to the TX	radians
R_t	Distance from TX (source transmitter) to PRA	meters
R_r	Distance from PRA to RX (receiver)	meters
z_s	Altitude (z-coordinate) of the array center point	meters
x_s	x-coordinate of the array center point	meters
y_s	y-coordinate of the array center point	meters
d	$\sqrt{x_s^2 + y_s^2}$	meters
(θ_0, ϕ_0)	Desired main-beam direction (relative to the PRA)	radians
$(\theta_{\text{null}}, \phi_{\text{null}})$	Desired null direction (relative to the PRA)	radians

3.2. Beamfocusing Condition

There exist necessary and sufficient conditions on the element positions $\mathbf{r}'_n$ that guarantee constructive interference of the scattered fields, thereby maximizing the radiated power in a prescribed direction. From Eq. (7), it follows that the scattered power in the direction (θ_0, ϕ_0) is maximized precisely when the contributions from all N elements are in phase. Since one element's phase can be fixed to zero via translation invariance (see Section 3.8) we can say that the radiation intensity in the direction (θ_0, ϕ_0) is maximal if and only if

$$-(\mathbf{k}(\theta_0, \phi_0) - \mathbf{k}(\theta_s, \phi_s)) \cdot (x'_n, y'_n, z'_n) \equiv 0 \pmod{2\pi}, \quad 1 \leq n \leq N. \quad (9)$$

Solving Eq. (9) for z'_n yields

$$(\cos \theta_0 - \cos \theta_s) z'_n \equiv (\sin \theta_s \cos \phi_s - \sin \theta_0 \cos \phi_0) x'_n + (\sin \theta_s \sin \phi_s - \sin \theta_0 \sin \phi_0) y'_n \pmod{\lambda_0}. \quad (10)$$

Since the relay system geometry enforces $0 \leq \theta_s < \pi/2 \leq \theta_0 \leq \pi$, we have $\cos \theta_0 \leq 0 < \cos \theta_s$, and it follows that $\cos \theta_0 - \cos \theta_s \neq 0$. Thus, the altitude constraint guaranteeing maximal radiation in the direction (θ_0, ϕ_0) is given by

$$z'_n = \frac{A x'_n + B y'_n}{D} + m_n \frac{\lambda_0}{D}, \quad m_n \in \mathbb{Z}, \quad (11)$$

where

$$A = \sin \theta_s \cos \phi_s - \sin \theta_0 \cos \phi_0, \quad (12)$$

$$B = \sin \theta_s \sin \phi_s - \sin \theta_0 \sin \phi_0, \quad (13)$$

$$D = \cos \theta_0 - \cos \theta_s \neq 0. \quad (14)$$

In the sequel when we impose this beamfocusing condition (11), we will often refer to it as the “fixed- z case” for short.

It is crucial that the altitudes z'_n prescribed by Eq. (11) remain bounded, since impractically large altitudes yield infeasible array geometries and severe grating lobes. The denominator, D , in Eq. (11) is small exactly when $\theta_s \approx \theta_0$, which requires both angles, θ_s and θ_0 , to be close to $\pi/2$; this occurs when $d \gg z_s$ in Fig. 2. Thus, the altitude required by Eq. (11) stays bounded as $\theta_s \rightarrow \theta_0$ only if the numerator $Ax'_n + By'_n$ stays small for every n . If the horizontal layout (x'_n, y'_n) is otherwise arbitrary – i.e., if the swarm contains at least two elements whose positions are linearly independent, then requiring $Ax'_n + By'_n \approx 0$ at two independent points forces A and B to vanish independently, i.e. $\cos \phi_s = \cos \phi_0$ and $\sin \phi_s = \sin \phi_0$, hence $\phi_0 \equiv \phi_s \pmod{2\pi}$. Physically when $\theta_s \approx \theta_0$, a swarm with a generic (non-collinear) 2D layout can only relay along the same azimuthal line as the incoming signal. That is, the transmitter, swarm, and receiver must lie in a common vertical plane, and the swarm cannot redirect the signal to a different azimuth.

With a properly chosen (x'_n, y'_n) , azimuthal redirection is nonetheless still possible when $\theta_s \approx \theta_0 \approx \pi/2$. At $D = 0$, Eq. (10) reduces to

$$0 \equiv Ax'_n + By'_n \pmod{\lambda_0}, \quad (15)$$

so the beamfocusing phase condition stops depending on the altitudes z'_n entirely in this regime.

Provided $\phi_0 \neq \phi_s$, we have that $(A, B) \neq (0, 0)$, so Eq. (15) always has a nontrivial line of solutions: elements placed on this line achieve exact coherent addition toward (θ_0, ϕ_0) , for any desired azimuth $\phi_0 \neq \phi_s$.

If $B \neq 0$, Eq. (15) can be solved for y'_n :

$$y'_n \equiv -\frac{A}{B} x'_n \pmod{\lambda_0 B^{-1}}. \quad (16)$$

If instead $B = 0$, then $A \neq 0$, and we solve for x'_n instead:

$$x'_n \equiv -\frac{B}{A} y'_n = 0 \pmod{\lambda_0 A^{-1}}. \quad (17)$$

Together, Eqs. (16) and (17) cover every case when $\phi_0 \neq \phi_s$.

For use in practice, Eq. (16) admits a convenient closed form. Sum-to-product identities give

$$y'_n \equiv \tan\left(\frac{\phi_s + \phi_0}{2}\right) x'_n \pmod{\lambda_0(\sin \theta_0)^{-1}(\sin \phi_s - \sin \phi_0)^{-1}}, \quad \phi_s \neq \phi_0 \pmod{2\pi}. \quad (18)$$

Taking the further approximation $\theta_0 \approx \pi/2 \Rightarrow \sin \theta_0 \approx 1$ simplifies the modulus and yields

$$y'_n \approx \tan\left(\frac{\phi_s + \phi_0}{2}\right) x'_n \pmod{\lambda_0(\sin \phi_s - \sin \phi_0)^{-1}}. \quad (19)$$

Note that Eq. (19) expresses y'_n as a function of x'_n , which becomes numerically awkward as the slope steepens, i.e., $B = \sin \phi_s - \sin \phi_0 \rightarrow 0$, which occurs when $\phi_s + \phi_0 \rightarrow (2l - 1)\pi$ for integer l (incident and outgoing azimuths nearly symmetric about the y -axis). This is not a physical obstruction, only a coordinate artifact of solving for y'_n rather than x'_n . The fix is to swap the dependent variable and use Eq. (17) in this regime.

3.3. Null Power Functional

To enforce a deep null at a specified direction $(\theta_{\text{null}}, \phi_{\text{null}})$, we define the null power functional:

$$N(\mathbf{x}) := U(\mathbf{x}, (\theta_{\text{null}}, \phi_{\text{null}})). \quad (20)$$

The optimization requires $N(\mathbf{x}) \leq \delta_{\text{null}}$ for a small enough threshold $\delta_{\text{null}} > 0$. In the fixed- z case, setting $\delta_{\text{null}} = (10^{-2}e(\theta_0)N)^2$ ensures that the radiated power in the direction $(\theta_{\text{null}}, \phi_{\text{null}})$ remains below -40 dB relative to the peak.

To ensure the problem is well-posed, the null direction, $(\theta_{\text{null}}, \phi_{\text{null}})$, must be distinct from both the main-beam direction (θ_0, ϕ_0) as well as the source direction (θ_s, ϕ_s) , because simultaneous focusing and nulling in the same direction is impossible. Notice beamfocusing in the direction (θ_s, ϕ_s) is guaranteed since evaluating $(\theta, \phi) = (\theta_s, \phi_s)$ in Eq. (7) yields $\mathbf{k}_{\text{in}} = \mathbf{k}_{\text{out}}$. Thus, it is advisable to choose ϕ_s and ϕ_0 such that their angular distance is less than half the main beamwidth to avoid grating lobes.

3.4. Spacing Constraints

To mitigate mutual coupling between scattering elements, we require an inter-scatterer distance of at least a quarter wavelength. Specifically, we enforce the minimum spacing condition

$$\|\mathbf{r}'_n(\mathbf{x}) - \mathbf{r}'_m(\mathbf{x})\| \geq d_{\min} = \frac{\lambda_0}{4} \quad \forall n \neq m.$$

To quantify deviations from this geometric threshold, we define the pairwise violation function

$$C_{n,m}(\mathbf{x}) := d_{\min} - \|\mathbf{r}'_n(\mathbf{x}) - \mathbf{r}'_m(\mathbf{x})\|,$$

and the corresponding aggregate penalty function

$$C(\mathbf{x}) := \sum_{n < m} \max\{0, C_{n,m}(\mathbf{x})\}.$$

Thus, the spacing constraint is satisfied exactly when

$$C_{n,m}(\mathbf{x}) \leq 0 \quad \forall n \neq m \iff C(\mathbf{x}) = 0.$$

3.5. Optimization Formulation

Let $\mathcal{X} = \mathbb{R}^{3N}$ in the free- z case, or $\mathcal{X} = \mathbb{R}^{2N}$ when the z -coordinates are fixed by the beamfocusing condition Eq. (11). The constrained optimization problem is given by

$$\begin{aligned} & \min_{\mathbf{x} \in \mathcal{X} \cap B_R} S(\mathbf{x}) \\ & \text{subject to } N(\mathbf{x}) \leq \delta_{\text{null}}, \\ & \quad C(\mathbf{x}) = 0, \\ & \quad U(\mathbf{x}, (\theta_0, \phi_0)) \geq p_{\min}. \end{aligned} \tag{21}$$

where $B_R \subset \mathcal{X}$ is the closed ball of radius R centered at the origin, $N(\mathbf{x}) \leq \delta_{\text{null}}$ ($\delta_{\text{null}} > 0$) enforces a null depth requirement, $C(\mathbf{x}) = 0$ enforces the minimum inter-element spacing, and for a given minimal power level $p_{\min}$, we require $U(\mathbf{x}, (\theta_0, \phi_0)) \geq p_{\min}$ to ensure the power in the main-beam direction is at least $p_{\min}$.

The cost functional S will be defined to penalize energy outside the main beam. We present two distinct formulations for S , the notch-window functional W and the peak sidelobe functional PS , defined in Sections 3.6.1 and 3.6.2 respectively. Note that in the fixed- z case, maximal main-beam power is guaranteed by construction, see Eq. (11). Thus, in the fixed- z case, the main-beam power constraint is automatically satisfied by construction, provided $p_{\min} < |e(\theta_0)|^2 N^2$.

The restriction to B_R serves two purposes. First, it ensures the far-field approximation underlying U remains valid. Second, it is without loss of generality: as shown in Section 3.8, both the cost S and the feasibility constraints are invariant under global translations of the array. Therefore, the only way to leave B_R is for the array diameter to grow without bound. Empirically, highly dispersed array configurations produce increasing sidelobe levels, so any minimizer of S over $\mathcal{X}$ should be attained within a sufficiently large B_R .

3.5.1. Existence of Global Minimizer

The feasible set, $\mathcal{F}$, of the optimization problem (21) can be expressed as

$$\mathcal{F} = C^{-1}(\{0\}) \cap N^{-1}([0, \delta_{\text{null}}]) \cap P^{-1}([p_{\min}, |e(\theta_0)|^2 N^2]) \cap B_R, \tag{22}$$

where $P(\mathbf{x}) := U(\mathbf{x}, (\theta_0, \phi_0))$.

Since C , N , and P are continuous, each preimage above is closed, and hence $\mathcal{F}$ is closed. As $\mathcal{F} \subset B_R$ and B_R is bounded, it follows that $\mathcal{F}$ is compact.

Moreover, $\mathcal{F}$ can be verified to be nonempty by construction of an explicit point $\bar{\mathbf{x}} \in \mathcal{F}$. First, choose the z'_n values of $\bar{\mathbf{x}}$ to satisfy the beamfocusing condition Eq. (11). We set

$$z'_n = \frac{A x'_n + B y'_n}{D},$$

where A, B, D are defined in Eqs. (12) to (14). Since z'_n satisfies the beamfocusing condition exactly, $P(\bar{\mathbf{x}}) = N^2 |e(\theta_0)|^2 \geq p_{\min}$, thus one of the constraints is satisfied.

For brevity let $\hat{k}_0, \hat{k}_s$, and $\hat{k}_{\text{null}}$ denote the unit directional vectors corresponding to the main-beam direction (θ_0, ϕ_0) , source (θ_s, ϕ_s) , and null $(\theta_{\text{null}}, \phi_{\text{null}})$ directions, respectively. Also let $(\Delta k_x, \Delta k_y, \Delta k_z) := \mathbf{k}(\theta_{\text{null}}, \phi_{\text{null}}) - \mathbf{k}(\theta_s, \phi_s)$. Expressing z'_n in terms of x'_n and y'_n yields

$$(\mathbf{k}(\theta_{\text{null}}, \phi_{\text{null}}) - \mathbf{k}(\theta_s, \phi_s)) \cdot \mathbf{r}'_n = Q x'_n + P y'_n. \quad (23)$$

where we define $Q = \Delta k_x + \Delta k_z A/D$ and $P = \Delta k_y + \Delta k_z B/D$.

We now claim that $(Q, P) \neq (0, 0)$. Suppose otherwise. Setting $\alpha := (\cos \theta_{\text{null}} - \cos \theta_s) / (\cos \theta_0 - \cos \theta_s)$ one can verify that $(Q, P) = (0, 0)$ if and only if

$$\hat{k}_{\text{null}} = (1 - \alpha) \hat{k}_s + \alpha \hat{k}_0. \quad (24)$$

Taking the squared magnitude of both sides gives

$$1 = |(1 - \alpha) \hat{k}_s + \alpha \hat{k}_0|^2 \implies 2\alpha(\alpha - 1)(1 - \hat{k}_s \cdot \hat{k}_0) = 0,$$

so either $\alpha = 0$, $\alpha = 1$, or $\hat{k}_s = \hat{k}_0$. However, none of these cases is permissible under our standing assumptions. Substituting into Eq. (24):

- (i) $\alpha = 0$ gives $\hat{k}_{\text{null}} = \hat{k}_s$, i.e. $(\theta_{\text{null}}, \phi_{\text{null}}) = (\theta_s, \phi_s)$, contradicting the standing assumption that the null direction differs from the source direction.
- (ii) $\alpha = 1$ gives $\hat{k}_{\text{null}} = \hat{k}_0$, i.e. $(\theta_{\text{null}}, \phi_{\text{null}}) = (\theta_0, \phi_0)$, contradicting the standing assumption that the null direction differs from the main-beam direction.
- (iii) $\hat{k}_s = \hat{k}_0$ means $(\theta_s, \phi_s) = (\theta_0, \phi_0)$, contradicting the standing assumption $\theta_s \neq \theta_0$.

Hence, $(Q, P) \neq (0, 0)$, as claimed.

The construction therefore splits into two cases:

- (i) $P \neq 0$. Define, for $n = 1, \dots, N$,

$$x'_n := d_{\min} n, \quad y'_n := \frac{1}{P} \left[\frac{2\pi(n-1)}{N} - Q d_{\min} n \right], \quad z'_n := \frac{A x'_n + B y'_n}{D}. \quad (25)$$

- (ii) $P = 0, Q \neq 0$. Define, for $n = 1, \dots, N$,

$$x'_n := \frac{2\pi(n-1)}{NQ}, \quad y'_n := d_{\min} n, \quad z'_n := \frac{A x'_n + B y'_n}{D}. \quad (26)$$

In both cases, either $x'_n = d_{\min} n$ or $y'_n = d_{\min} n$, which is enough to satisfy the minimal spacing constraint. Thus, $C(\bar{\mathbf{x}}) = 0$. By Eq. (23) and the definitions (25) and (26), direct substitution gives

$$Q x'_n + P y'_n = \frac{2\pi(n-1)}{N} \quad \text{for } n = 1, \dots, N,$$

in both cases. Therefore the array factor at the null direction is

$$\sum_{n=1}^N \exp(-i \Delta \mathbf{k}_{\text{null}} \cdot \mathbf{r}'_n) = \sum_{n=1}^N \exp(-2\pi i(n-1)/N) = 0,$$

as the sum of the roots of unity of order N vanishes. Hence,

$$N(\bar{\mathbf{x}}) = U(\bar{\mathbf{x}}, (\theta_{\text{null}}, \phi_{\text{null}})) = |e(\theta_{\text{null}})|^2 \cdot 0 = 0 \leq \delta_{\text{null}}.$$

For R chosen sufficiently large, $\bar{\mathbf{x}} \in B_R$, and thus we can conclude that $\bar{\mathbf{x}} \in \mathcal{F}$.

Remark 1. The explicit construction above shows that the feasible set is nonempty. Furthermore, if all inequality constraints discussed above are strictly satisfied (i.e. if $p_{\min} \neq N^2|e(\theta_0)|^2$, $\delta_{\text{null}} \neq 0$), the configuration remains in the feasible set even under small perturbations.

We now state a general existence result.

Theorem 1. *Let $\mathcal{F} \subset \mathcal{X}$ be the feasible set defined above. If $S : \mathcal{X} \rightarrow \mathbb{R}$ is lower semicontinuous, then the optimization problem Eq. (21) admits a global minimizer; that is, there exists $\mathbf{x}^* \in \mathcal{F}$ such that*

$$S(\mathbf{x}^*) = \min_{\mathbf{x} \in \mathcal{F}} S(\mathbf{x}).$$

PROOF. Since $\mathcal{F}$ is a compact, nonempty, and S is lower semicontinuous, the claim follows from the Weierstrass Extreme Value Theorem.

3.6. Sidelobe Control Functionals

As mentioned in Section 3.5, we wish to design a cost functional S that minimizes radiated energy outside a designated main-beam region. While an indirect method like maximizing directivity concentrates energy in or near the focus direction, it does not explicitly prevent the formation of narrow, high-intensity sidelobes outside the main-beam region, nor does it strictly constrain the main beamwidth. As is well established in antenna array literature, fundamental trade-offs exist between minimizing sidelobe levels, minimizing main beamwidth, and maximizing directivity: optimizing one inherently works against the others at a global level.

Given the application considered in this paper, where low probability of detection or interference is the goal, and sufficient power delivery is already guaranteed by applying the beamfocusing condition (11), our primary optimization goals are strictly to suppress sidelobes while simultaneously constraining the main beam to an acceptable angular width. To achieve this, we introduce two distinct objective functionals, each differently navigating the challenges of sidelobe suppression.

Ideally, one would directly minimize the exact Peak Sidelobe Level (PSL), defined as the global maximum of the radiation pattern outside the main beam. However, tracking sidelobes (local maxima) with respect to changing array parameters poses significant theoretical and numerical hurdles. While tracking the global maximum (the main beam) is continuous (as it involves maximizing a continuous function over a fixed compact set), tracking local maxima relies on the Implicit Function Theorem and strictly requires critical points to remain non-degenerate. If a perturbation in the array geometry causes a critical point to become degenerate, the optimization landscape undergoes a bifurcation. For instance, a single flat peak might abruptly split into two distinct peaks with a new local minimum emerging between them. This bifurcation severely disrupts gradient-based optimization algorithms due to three primary factors:

- a) Discontinuous Cardinality: The number of critical points (sidelobes) in the spatial domain jumps discontinuously.
- b) Loss of Sidelobe Identity: As a peak splits, the continuous identity of the tracked sidelobe becomes ill-defined.
- c) Non-Differentiable Landscape: The global maximum of the sidelobes can abruptly transition from one local peak to another, creating sharp, non-differentiable points in the objective function.

Consequently, relying on a direct measure of the exact PSL requires either proving that the set of degenerate configurations is of measure zero—ensuring the optimizer almost never encounters them—or adopting a more forgiving surrogate functional. We explore both avenues below.

First, in Section 3.6.1, we introduce an *Exclusion Rectangle* (or *Notch Window*) functional. This objective prescribes a fixed exclusion region, Ω_{SL} , outside the intended main beam and seeks to minimize the L^∞ -norm of the radiation pattern over this region. This guarantees continuity and avoids the bifurcation issues entirely. However, if the main beam broadens and spills into Ω_{SL} , the functional penalizes this spillover as if it were a sidelobe. While this usefully forces the optimizer to constrain the main beamwidth, it inherently creates a multiobjective ambiguity where the functional conflates main-beam spillover with true sidelobe peaks. To resolve this ambiguity, in Section 3.6.2 we formalize a precise *Peak Sidelobe Functional*. This formulation strictly tracks the non-main-beam critical points of the radiation pattern, treating the exact PSL problem directly, but requires additional non-degeneracy assumptions to ensure the necessary continuity properties.

3.6.1. Exclusion Rectangle (Notch Window) Functional

We define an exclusion, or *notch*, window $w : \Omega \rightarrow \{0, 1\}$ by

$$w(\theta, \phi) = \begin{cases} 0, & \text{if } |\theta - \theta_0| < M_\theta \text{ and } |\phi - \phi_0| < M_\phi, \\ 1, & \text{otherwise.} \end{cases} \quad (27)$$

The region that w maps to 0 is exactly a rectangle centered at (θ_0, ϕ_0) (corresponding to the intended main-beam direction), of dimensions $2M_\theta$ (in elevation) and $2M_\phi$ (in azimuth). Here, M_θ must be in $(0, \pi/2)$ and $M_\phi \in (0, \pi)$. Thus, let $\Omega_{\text{SL}} = w^{-1}(\{1\})$ denote the main-beam exclusion region.

We then define the functional $W : \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ by

$$W(\mathbf{x}) = \max_{(\theta, \phi) \in \Omega} w(\theta, \phi) U(\mathbf{x}, (\theta, \phi)) = \max_{(\theta, \phi) \in \Omega_{\text{SL}}} U(\mathbf{x}, (\theta, \phi)). \quad (28)$$

Continuity of W on $\mathcal{X}$ follows from the Berge Maximum Theorem, since U is continuous in both arguments and $w^{-1}(\{1\})$ is compact.

For optimization restricted to the θ_0 azimuthal plane we take

$$W^{(2D)}(\mathbf{x}) := \max_{(\theta_0, \phi) \in \Omega} w(\theta_0, \phi) U(\mathbf{x}, (\theta_0, \phi)). \quad (29)$$

Corollary 2. *Let $S(\mathbf{x}) = W(\mathbf{x})$, where W is defined in Eq. (28). Then the optimization problem (21) admits a global minimizer.*

PROOF. As shown above, W is continuous on $\mathcal{X}$, and hence lower semicontinuous. The result follows from Theorem 1.

3.6.2. Peak Sidelobe Functional

We define a functional that maps an array geometry $\mathbf{x} \in \mathcal{X}$ to the peak sidelobe level of its associated radiation pattern $U(\mathbf{x}, \cdot)$, and establish conditions under which it is continuous. Throughout this section we assume the z -coordinates are determined by the beamfocusing condition (11), so the decision space is $\mathcal{X} = \mathbb{R}^{2N}$. The angular domain is taken to be

$$\Omega := [\theta_0 - \pi/2, \theta_0 + \pi/2] \times [\phi_0 - \pi, \phi_0 + \pi],$$

reduced to $\Omega = \{\theta_0\} \times [\phi_0 - \pi, \phi_0 + \pi]$ (or equivalently $\Omega = [\phi_0 - \pi, \phi_0 + \pi]$) when restricting to a 2D cut of the pattern.

By construction, the main beam is steered toward (θ_0, ϕ_0) , so

$$\max_{(\theta, \phi) \in \Omega} e(\theta)^{-2} \cdot U(\mathbf{x}, (\theta, \phi)) = e(\theta_0)^{-2} \cdot U(\mathbf{x}, (\theta_0, \phi_0)) \quad \text{for all } \mathbf{x} \in \mathcal{X}.$$

We isolate the sidelobe peak points by defining the correspondence $\Gamma_\varepsilon : \mathcal{X} \rightrightarrows \Omega$,

$$\Gamma_\varepsilon(\mathbf{x}) := \{(\theta, \phi) \in \Omega \mid \nabla_\Omega U(\mathbf{x}, (\theta, \phi)) = \mathbf{0}\} \setminus B_\varepsilon((\theta_0, \phi_0)), \quad (30)$$

which collects the critical points of $U(\mathbf{x}, \cdot)$ outside an ε -ball around the main beam. We augment this with the set $\Omega_{\text{SL}} := w^{-1}(\{1\})$ (as defined in the preceding section) by setting

$$\Gamma(\mathbf{x}) := \Gamma_\varepsilon(\mathbf{x}) \cup \Omega_{\text{SL}}, \quad (31)$$

which ensures that $\Gamma(\mathbf{x})$ is always nonempty. Note that Ω_{SL} can be arbitrarily small in measure (although nonzero). Then we define the peak sidelobe functional as

$$PS(\mathbf{x}) := \max_{(\theta, \phi) \in \Gamma(\mathbf{x})} U(\mathbf{x}, (\theta, \phi)). \quad (32)$$

To apply the Berge Maximum Theorem to Eq. (32), we must establish that Γ is nonempty, compact-valued, and hemicontinuous. The first two properties hold globally while lower hemicontinuity requires a restriction of $\mathcal{X}$ to exclude degenerate critical points, which we now introduce.

Let $\tilde{\Omega} = \Omega \setminus B_\epsilon((\theta_0, \phi_0))$ and let $H(\mathbf{x}, (\theta, \phi)) = \nabla_\Omega^2 U(\mathbf{x}, (\theta, \phi))$ denote the 2×2 Hessian of U with respect to (θ, ϕ) . Note that U is smooth and H is symmetric. For $\kappa > 0$ define

$$X_{\text{robust}}(\kappa) = \{\mathbf{x} \in \mathcal{X} : \|\nabla_\Omega U(\mathbf{x}, \omega)\| \geq \kappa \quad \text{for } \omega \in \partial B_\epsilon((\theta_0, \phi_0))\}. \quad (33)$$

This set excludes array configurations whose main beam has plateaus around the focus direction, ridges or nearly flat saddles extending outward from the focus direction, or the focus direction is a weakly curved maximum where gradients become arbitrarily small nearby. Then, for a fixed $\xi > 0$, define

$$X_0(\xi, \kappa) = \left\{ x \in X_{\text{robust}}(\kappa) \mid \|\nabla_\Omega U(x, y)\| + \sigma_{\min}(H(x, y)) \geq \xi \quad \forall y \in \tilde{\Omega} \right\}, \quad (34)$$

where $\sigma_{\min}$ denotes the smallest singular value. This set in turn also excludes array configurations that produce degenerate or near-degenerate sidelobe critical points.

Note that by the compactness of $\partial B_\epsilon(\theta_0, \phi_0)$ and continuity of $\|\nabla_\Omega U\|$ it follows that $X_{\text{robust}}(\kappa)$ is closed in $\mathcal{X}$. Also, by the compactness of $\tilde{\Omega}$, and continuity of $\|\nabla_\Omega U\| + \sigma_{\min}(H)$, $X_0(\xi, \kappa)$ is closed in $X_{\text{robust}}(\kappa)$. Thus, $X_0(\xi, \kappa)$ is closed in $\mathcal{X}$.

Lemma 3. *The correspondence¹ $\Gamma : \mathcal{X} \rightrightarrows \Omega$ satisfies the following:*

- (i) $\Gamma(\mathbf{x})$ is nonempty for all $\mathbf{x} \in \mathcal{X}$.
- (ii) $\Gamma(\mathbf{x})$ is compact for all $\mathbf{x} \in \mathcal{X}$.
- (iii) Γ is upper hemicontinuous on $\mathcal{X}$.
- (iv) Γ is lower hemicontinuous on $X_0(\xi, \kappa)$.

PROOF. (i) Immediate, since $\Omega_{\text{SL}} \subseteq \Gamma(\mathbf{x})$ and Ω_{SL} is nonempty.

- (ii) By definition, $\Omega_{\text{SL}} = \omega^{-1}(\{1\})$, which is the complement of the open rectangle in Ω . Hence Ω_{SL} is closed in Ω . Since Ω is compact, it follows that Ω_{SL} is compact.

Next, consider

$$\Gamma_\epsilon(\mathbf{x}) = \{(\theta, \phi) \in \Omega \mid \nabla_\Omega U(\mathbf{x}, (\theta, \phi)) = \mathbf{0}\} \setminus B_\epsilon((\theta_0, \phi_0)).$$

Since U is continuously differentiable, $\nabla_\Omega U(\mathbf{x}, (\theta, \phi))$ is continuous in (θ, ϕ) . It follows that the set

$$\{(\theta, \phi) \in \Omega \mid \nabla_\Omega U(\mathbf{x}, (\theta, \phi)) = \mathbf{0}\}$$

is closed in Ω as the preimage of $\{\mathbf{0}\}$ under a continuous mapping. Moreover, $B_\epsilon((\theta_0, \phi_0))$ is open, so its complement is closed. Hence $\Gamma_\epsilon(\mathbf{x})$ is the intersection of two closed subsets of Ω , and is therefore closed in Ω . Since Ω is compact, it follows that $\Gamma_\epsilon(\mathbf{x})$ is compact.

Finally, $\Gamma(\mathbf{x}) = \Gamma_\epsilon(\mathbf{x}) \cup \Omega_{\text{SL}}$ is the union of two compact sets, and is therefore compact.

- (iii) We use the sequential characterization of upper hemicontinuity. Let $\mathbf{x}_k \rightarrow \mathbf{x}^*$ in $\mathcal{X}$ and suppose $(\theta_k, \phi_k) \in \Gamma(\mathbf{x}_k)$ with $(\theta_k, \phi_k) \rightarrow (\theta^*, \phi^*)$. If infinitely many terms, (θ_k, ϕ_k) , lie in Ω_{SL} , then $(\theta^*, \phi^*) \in \Omega_{\text{SL}} \subseteq \Gamma(\mathbf{x}^*)$ due to Ω_{SL} being closed and thus, the proof is complete.

Otherwise, there is a subsequence $\{n_k\}$ such that $(\theta_{n_k}, \phi_{n_k}) \in \Gamma_\epsilon(\mathbf{x}_{n_k})$ for every k , so that $\nabla_\Omega U(\mathbf{x}_{n_k}, (\theta_{n_k}, \phi_{n_k})) = \mathbf{0}$. By continuity of $\nabla_\Omega U$, taking $k \rightarrow \infty$ gives $\nabla_\Omega U(\mathbf{x}^*, (\theta^*, \phi^*)) = \mathbf{0}$. Since each $(\theta_k, \phi_k) \notin B_\epsilon((\theta_0, \phi_0))$ and this ball is open, the limit satisfies $(\theta^*, \phi^*) \notin B_\epsilon((\theta_0, \phi_0))$. Hence $(\theta^*, \phi^*) \in \Gamma_\epsilon(\mathbf{x}^*) \subseteq \Gamma(\mathbf{x}^*)$, as required.

- (iv) We use the sequential characterization of lower hemicontinuity. Let $x_0 \in X_0(\xi, \kappa)$ and $y_0 \in \Gamma(x_0)$. Let $\{x_k\} \subset X_0(\xi, \kappa)$ be any sequence such that $x_k \rightarrow x_0$. We must show there exists a sequence $y_k \in \Gamma(x_k)$ such that $y_k \rightarrow y_0$.

Because $\Gamma(x_0) = \Gamma_\epsilon(x_0) \cup \Omega_{\text{SL}}$, we proceed by analyzing two cases for y_0 :

- (a) $y_0 \in \Omega_{\text{SL}}$. Since set Ω_{SL} is a fixed region independent of the configuration x for any k , $y_0 \in \Gamma(x_k)$. We simply define the constant sequence $y_k = y_0$. Trivially, $y_k \in \Gamma(x_k)$ and $y_k \rightarrow y_0$.

¹See Section B for a further discussion of correspondences & the Berge Maximum Theorem.

(b) $y_0 \in \Gamma_\epsilon(x_0)$. By definition, $y_0 \in \tilde{\Omega}$ and $\nabla_\Omega U(x_0, y_0) = \mathbf{0}$. Because $x_0 \in X_0(\xi, \kappa)$, we have that

$$\|\nabla_\Omega U(x_0, y_0)\| + \sigma_{\min}(H(x_0, y_0)) \geq \xi.$$

Since the gradient norm is zero, this implies $\sigma_{\min}(H(x_0, y_0)) \geq \xi > 0$. Thus, the Hessian $H(x_0, y_0)$ is invertible.

Define the gradient mapping $F(x, y) = \nabla_\Omega U(x, y)$. We have $F(x_0, y_0) = \mathbf{0}$ and its partial Jacobian $D_y F(x_0, y_0) = H(x_0, y_0)$ is non-singular. By the Implicit Function Theorem, there exist open neighborhoods $\mathcal{N}(x_0) \subset \mathcal{X}$ of x_0 and $\mathcal{W}(y_0) \subset \Omega$ of y_0 , along with a unique continuous mapping $y : \mathcal{N}(x_0) \rightarrow \mathcal{W}(y_0)$, such that $y(x_0) = y_0$ and $F(x, y(x)) = \mathbf{0}$ for all $x \in \mathcal{N}(x_0)$.

Because $x_k \rightarrow x_0$, there exists an integer K such that for all $k \geq K$, $x_k \in \mathcal{N}(x_0)$. For these k , we define $y_k = y(x_k)$. By the continuity of the implicit function, y , as $x_k \rightarrow x_0$, we have $y_k \rightarrow y_0$. Note that as established before $F(x_0, y_0) = \mathbf{0}$. But, since $x_0 \in X_0(\xi, \kappa) \subset X_{\text{robust}}(\kappa)$, we also have that $\|F(x_0, \omega)\| \geq \kappa > 0$ for all $\omega \in \partial B_\epsilon((\theta_0, \phi_0))$. Hence, it must be that $y_0 \notin \partial B_\epsilon((\theta_0, \phi_0))$. So, y_0 is strictly in the interior of $\tilde{\Omega}$, and therefore the sequence y_k will eventually reside entirely within $\tilde{\Omega}$. This, combined with the fact that $F(x_k, y_k) = \mathbf{0}$ lets us conclude that $y_k \in \Gamma_\epsilon(x_k) \subset \Gamma(x_k)$ for sufficiently large k .

Theorem 4. *The peak sidelobe functional $PS : \mathcal{X} \rightarrow \mathbb{R}$, as defined in Eq. (32), is continuous on the set X_0 specified in Eq. (34).*

PROOF. It follows from Lemma 3 that Γ is nonempty, compact-valued, upper and lower hemicontinuous on $X_0(\xi, \kappa)$, i.e. Γ is continuous there. Since $U(\mathbf{x}, (\theta, \phi))$ is jointly continuous in $\mathbf{x}$ and (θ, ϕ) , part (iii) of the Berge Maximum Theorem (Theorem 7) applies directly to Eq. (32), yielding continuity of PS on $X_0(\xi, \kappa)$.

On X_0 define the feasible set $\mathcal{F}_0$ of PS by

$$\mathcal{F}_0 := C^{-1}(\{0\}) \cap N^{-1}([0, \delta_{\text{null}}]) \cap P^{-1}([p_{\min}, |e(\theta_0)|^2 N^2]) \cap X_0(\xi, \kappa) \cap B_R = \mathcal{F} \cap X_0(\xi, \kappa). \quad (35)$$

Our preliminary numerical investigations suggest the following conjecture:

Conjecture 5. *For appropriately chosen $\xi, \kappa, \epsilon > 0$, the set $\mathcal{X} \setminus X_0$ is nowhere dense in $\mathcal{X}$. Then it follows immediately as a consequence of Remark 1 that the feasible set, $\mathcal{F}_0$, is nonempty.*

Corollary 6. *Assume that $\mathcal{F}_0 = \mathcal{F} \cap X_0(\xi, \kappa)$ is nonempty. Then the optimization problem (21) with $S(\mathbf{x}) = PS(\mathbf{x})$ admits a global minimizer on $\mathcal{F}_0$.*

PROOF. By Theorem 4, PS is continuous (hence lower semicontinuous) on X_0 . The feasible set $\mathcal{F}_0$ is compact as the intersection of closed sets with the compact ball B_R . Furthermore, $\mathcal{F}_0$ is nonempty by assumption. The conclusion follows from the Weierstrass Extreme Value Theorem.

The nonemptiness assumption is natural in the present setting because degenerate critical configurations are expected to form a small exceptional set. A rigorous proof of this statement is beyond the scope of the present paper.

3.7. Differentiable Surrogates

Notice that the functionals, W and C , in the previous sections, are not differentiable due to the presence of the max operator. To enable the use of derivative-based optimization methods, we introduce smooth surrogate functions whose gradients and Hessians can be computed analytically.

We define

$$W_{p_1}(\mathbf{x}) = \frac{1}{p_1} \log \left(\int_{\Omega_{\text{SL}}} \exp(p_1 \cdot U(\mathbf{x}, (\theta, \phi))) d(\theta, \phi) \right) \rightarrow W(\mathbf{x}) \text{ as } p_1 \rightarrow \infty. \quad (36)$$

Here we employed a continuous analogue of the well-known LogSumExp operator, defined for a vector $\mathbf{v} \in \mathbb{R}^n$, as

$$\text{LSE}_{p_1}(\mathbf{v}) := \frac{1}{p_1} \log \sum_{i=1}^n \exp(p_1 v_i), \quad \mathbf{v} = (v_1, v_2, \dots, v_n).$$

For the numerical implementation of W_{p_1} we take,

$$W_{p_1}(\mathbf{x}) := \frac{1}{p_1} \log \sum_{(\theta, \phi) \in \mathcal{G}} \exp(p_1 \cdot U(\mathbf{x}, (\theta, \phi))) = U_{\max} + \frac{1}{p_1} \log \sum_{(\theta, \phi) \in \mathcal{G}} \exp(p_1 \cdot (U(\mathbf{x}, (\theta, \phi)) - U_{\max})) \quad (37)$$

where $\mathcal{G} \subset \Omega_{\text{SL}}$ denotes a discrete angular grid used to mesh Ω_{SL} and $U_{\max}$ the maximum radiation intensity over $\mathcal{G}$. Although a L^p -norm can also be used as a smooth maximum, we found this formulation to be more numerically stable.

To regularize the spacing constraint we approximate $\max\{0, C_{n,m}(\mathbf{x})\}$ using the SoftPlus function:

$$\text{SoftPlus}_{p_2}(C_{n,m}(\mathbf{x})) = \frac{1}{p_2} \log(1 + \exp(p_2 \cdot C_{n,m}(\mathbf{x}))) \rightarrow \max\{0, C_{n,m}(\mathbf{x})\} \text{ as } p_2 \rightarrow \infty.$$

Then, we define

$$C_{p_2}(\mathbf{x}) := \sum_{n < m} \frac{1}{p_2} \log(1 + \exp(p_2 \cdot C_{n,m}(\mathbf{x}))) \rightarrow C(\mathbf{x}) \text{ as } p_2 \rightarrow \infty. \quad (38)$$

A concise derivation of the gradients and Hessians of C_{p_2} , N , and W_{p_1} is provided in Section C.

3.8. Translational Invariance of the Array Response

Let $\Delta \mathbf{k} = \mathbf{k}(\theta, \phi) - \mathbf{k}(\theta_s, \phi_s)$ for brevity. Then

$$U(\mathbf{x}, (\theta, \phi)) = \left| e(\theta, \phi) \sum_{n=1}^N \exp(-i \Delta \mathbf{k} \cdot \mathbf{r}'_n) \right|^2.$$

If we translate all elements by $\mathbf{t}$, i.e. $\mathbf{r}''_n = \mathbf{r}'_n + \mathbf{t}$, then

$$\sum_{n=1}^N \exp(-i \Delta \mathbf{k} \cdot \mathbf{r}''_n) = \exp(-i \Delta \mathbf{k} \cdot \mathbf{t}) \sum_{n=1}^N \exp(-i \Delta \mathbf{k} \cdot \mathbf{r}'_n),$$

so the magnitude squared of the above quantity (and hence U) is unchanged since $|\exp(-i \Delta \mathbf{k} \cdot \mathbf{t})| = 1$. Moreover, $\|\mathbf{r}''_n - \mathbf{r}''_m\| = \|\mathbf{r}'_n - \mathbf{r}'_m\|$, so spacing constraints are preserved. Thus any translated configuration is equally feasible and has the same U . Hence we fix one of our array elements to the center of the swarm, (x_s, y_s, z_s) , without loss of generality throughout our entire optimization scheme.

3.9. Constraint Handling Strategies

For unconstrained optimization methods we enforce constraints by reformulating the problem as an unconstrained penalized minimization. Specifically, we consider

$$\min_{\mathbf{x} \in \mathbb{R}^{3N}} f_{\mu}(\mathbf{x}) := S(\mathbf{x}) + \mu_1 N(\mathbf{x}) + \mu_2 C(\mathbf{x}), \quad (39)$$

where the penalty parameters $\mu_1, \mu_2 > 0$ control the strength of the constraint enforcement. In practice we set $\mu_1 = 1$ and $\mu_2 = 100$.

Alternatively, for constrained optimization methods we rely on the generalized Lagrangian formulation. Assuming we seek to minimize $S(\mathbf{x})$ subject to the inequality constraints $N(\mathbf{x}) - \delta_{\text{null}} \leq 0$ and $C_{n,m}(\mathbf{x}) \leq 0$, we define the Lagrangian as:

$$\mathcal{L}(\mathbf{x}, \lambda) = S(\mathbf{x}) + \lambda_0 (N(\mathbf{x}) - \delta_{\text{null}}) + \sum_{n < m} \lambda_{n,m} C_{n,m}(\mathbf{x}), \quad (40)$$

where λ is the vector of Karush-Kuhn-Tucker (KKT) multipliers. In this formulation, each spacing constraint $C_{n,m}$ is treated individually with its own non-negative multiplier $\lambda_{n,m}$, and λ_0 corresponds to the null constraint.

4. Optimization Algorithms

4.1. Ansatz initialization: Concentric Circular Array

We consider the following physically informed array configuration. The z coordinates are determined by the beamfocusing condition. The (x, y) coordinates of the elements are determined as follows: one element is fixed at the origin, and the remaining elements are arranged to lie on a pair of concentric circles: N_1 elements uniformly spaced on a ring of radius d_1 , and N_2 elements uniformly spaced on a ring of radius d_2 . The rings are azimuthally offset from each other by an angle of $\Delta \in [0, 2\pi)$.

That is, for

$$\alpha_m = \frac{2\pi(m-1)}{N_1} \quad (m = 1, \dots, N_1), \quad \beta_m = \frac{2\pi(m-1)}{N_2} \quad (m = 1, \dots, N_2),$$

the element positions are given by

$$\begin{aligned} \mathbf{r}'_m(d_1) &= (d_1 \cos \alpha_m, d_1 \sin \alpha_m, z_m), & 1 \leq m \leq N_1, \\ \mathbf{r}'_{m+N_1}(d_2, \Delta) &= (d_2 \cos(\beta_m + \Delta), d_2 \sin(\beta_m + \Delta), z_{m+N_1}), & 1 \leq m \leq N_2, \\ \mathbf{r}'_N &= (0, 0, z_N). \end{aligned} \quad (41)$$

We select the best (d_1, d_2, Δ) , through a direct grid search, to minimize a chosen cost function S . Concretely, we solve the discrete problem

$$(d_1^*, d_2^*, \Delta^*) \in \arg \min_{d_1, d_2 \in D, \Delta \in \mathcal{A}} S(\mathbf{x}(d_1, d_2, \Delta)), \quad (42)$$

where D and $\mathcal{A}$ are discretizations of the spaces $(0, \lambda_0)$ and $[0, 2\pi)$ respectively (λ_0 is the free-space wavelength). Notice here we don't yet worry about synthesizing a null (so it's not necessarily the case that $N(\mathbf{x}(d_1^*, d_2^*, \Delta^*)) \leq \delta_{\text{null}}$).

The full Concentric-ring (Ansatz) initialization procedure described in this section is summarized in [Algorithm 1](#) below. In particular, for the $N = 9$ case with $N_1 = N_2 = 4$, $\theta_0 = 90^\circ$, $\phi_0 = \phi_s = 45^\circ$, cost function $S = W^{2D}$ with parameter $M_\phi = 60^\circ$ this procedure selected $\Delta^* = 45^\circ$, $d_1^* = 0.48786$, and $d_2^* = 0.34374$, after which we use $\mathbf{x}(d_1^*, d_2^*, \Delta^*)$ as the initialization for subsequent optimizations. A visualization of this particular layout can be seen in [Fig. 3](#).

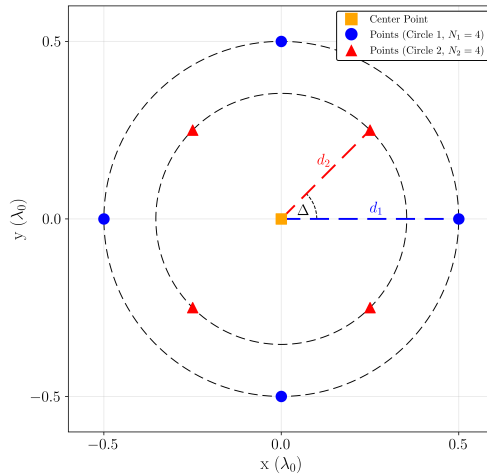


Fig. 3: Optimized element positions in the xy -plane (in units of wavelength), obtained by solving Eq. (42) for focus direction $\phi_0 = 45^\circ$. The cost function used is $S = W^{(2D)}$ with parameter $M_\phi = 60^\circ$.

Algorithm 1 Concentric-Ring (Ansatz) Initialization

Require: Total elements $N = 1 + N_1 + N_2$ (1 center, N_1 inner ring, N_2 outer ring);
 Radius grid $\mathcal{D} \subset (0, \lambda_0)$; Offset grid $\mathcal{A} \subset [0, 2\pi)$;
 Cost function $S(\mathbf{x})$

Ensure: Initial layout vector $\mathbf{x}_0$

```

1: Define uniform angular spacings:
    $\alpha_m \leftarrow 2\pi(m-1)/N_1$  for  $m = 1, \dots, N_1$ 
    $\beta_n \leftarrow 2\pi(n-1)/N_2$  for  $n = 1, \dots, N_2$ 
2: for all  $(d_1, d_2, \Delta) \in \mathcal{D} \times \mathcal{D} \times \mathcal{A}$  do ▷ Direct grid search
3:   Set center element position at  $(0, 0)$ 
4:   Set inner-ring positions:  $\mathbf{p}_m^{(1)} \leftarrow d_1(\cos \alpha_m, \sin \alpha_m)$ 
5:   Set outer-ring positions:  $\mathbf{p}_n^{(2)} \leftarrow d_2(\cos(\beta_n + \Delta), \sin(\beta_n + \Delta))$ 
6:   Calculate heights  $z_i$  via the beamfocusing condition
7:   Assemble the full layout vector  $\mathbf{x}(d_1, d_2, \Delta)$ 
8:   Evaluate configuration cost:  $S(\mathbf{x}(d_1, d_2, \Delta))$ 
9: end for
10:  $(d_1^*, d_2^*, \Delta^*) \leftarrow \arg \min_{(d_1, d_2, \Delta)} S(\mathbf{x}(d_1, d_2, \Delta))$ 
11: return  $\mathbf{x}_0 \leftarrow \mathbf{x}(d_1^*, d_2^*, \Delta^*)$ 
    
```

4.2. Optimization Algorithms

Starting from the physically informed concentric circular initialization described in Section 4.1, we evaluate several gradient-based local methods and benchmark their performance against an intensive global optimization strategy.

4.2.1. Local Descent Methods

We employ the following local optimization solvers:

(i) MATLAB® second-order solvers:

- `fmincon`: Interior-point method for constrained optimization.
- `fminunc`: Trust-region method for unconstrained optimization.

(ii) ADAM: A first-order gradient descent optimizer with adaptive moment estimation. We use standard momentum parameters $\beta_1 = 0.9$, $\beta_2 = 0.99$, and a learning rate $\alpha = 0.01$.

(iii) Exact TR (Exact Trust-Region): A custom trust-region algorithm that solves each trust-region subproblem exactly.

(iv) ZIPC: A Zoom-In Penalty Continuation scheme (built on top of Exact TR) that geometrically scales the null penalty weight to synthesize deep nulls without increasing the previously achieved sidelobe level.

For the unconstrained solvers (`fminunc`, ADAM, Exact TR, and ZIPC), we minimize the penalized aggregate functional f_μ defined in Eq. (39). In contrast, for the constrained solver `fmincon`, the inequality constraints are handled using its interior-point algorithm. The constraints are enforced internally via a logarithmic barrier function, solving a sequence of approximate equality-constrained problems that converge to a solution satisfying the KKT conditions of the generalized Lagrangian which we defined in Eq. (40).

4.2.2. Global Benchmarking

To provide a baseline for the performance of our local methods, we benchmark against a computationally intensive global strategy: `MultiStart` paired with `fminunc` as the local solver. We sample 10,000 initial configurations in the decision variable domain via Latin Hypercube Sampling and launch an independent `fminunc` descent from each point. The best final solution across all runs is retained.

The results of the `MultiStart` experiments underscore the extreme nonconvexity of the objective landscape. In the 2D azimuthal optimization, only 13.68% of the randomly sampled initializations converged to solutions with a peak sidelobe level below -20 dB, and less than 0.72% achieved levels below -25 dB. This sensitivity highlights the necessity of our physically informed concentric circular initialization chosen as the starting point.

4.3. Trust-Region Method with Exact Subproblem

To minimize the smooth objective $f : \mathbb{R}^{3N} \rightarrow \mathbb{R}$ defined in Eq. (39), we employ the exact trust-region framework of Conn et al. [6]. We solve the quadratic subproblem globally using the secular-equation approach of Moré and Sorensen [23]. At the current iterate $\mathbf{x}$ (with gradient $g = \nabla f(\mathbf{x})$ and Hessian $H = \nabla^2 f(\mathbf{x})$) the quadratic model $m(p) = g^\top p + 0.5 p^\top H p$ is minimized over the trust region $\|p\| \leq \Delta$; the step is accepted or rejected by the ratio of actual to predicted reduction, and Δ is adapted accordingly. The algorithm terminates when first-order stationarity is achieved and the local geometry is strictly convex, that is, when $\|g_k\| \leq \epsilon_{\text{grad}}$ and $\lambda_{\min}(H_k) > 0$. The full trust-region algorithm is described concisely in Algorithm 2.

We note a distinction between this exact approach and standard large-scale implementations, such as the trust-region algorithm in MATLAB's `fminunc`, which is also evaluated in this work. While `fminunc` approximates the subproblem solution by restricting the step to a two-dimensional subspace (spanned by the gradient and the Newton direction, or a direction of negative curvature), the exact method solves the full-dimensional subproblem globally. Although computationally more demanding per iteration due to the required eigensystem decomposition, it is demonstrated in the numerics section below that the exact method more effectively navigates the nonconvex landscape, converging to superior local minima in fewer iterations.

Algorithm 2 Trust-Region Minimization of f with Exact Subproblem

Require: Objective f ; Initial design $\mathbf{x}_0$;

Initial radius $\Delta_0 > 0$, max radius $\Delta_{\max}$, accept threshold $\eta \in (0, 1/4)$;

Tolerances $\epsilon_{\text{grad}} > 0$ and $\epsilon_{\text{nc}} > 0$

Ensure: Local minimizer $\mathbf{x}^*$ satisfying first and second-order stationarity conditions

```

1:  $\mathbf{x} \leftarrow \mathbf{x}_0$ 
2:  $\Delta \leftarrow \Delta_0$ 
3: repeat
4:   Compute gradient  $g \leftarrow \nabla f(\mathbf{x})$  and Hessian  $H \leftarrow \nabla^2 f(\mathbf{x})$ 
5:   Exact Subproblem (Moré–Sorensen): Compute step  $p$  globally:
      
$$p = \begin{cases} -H^{-1}g & \text{if } H > 0 \text{ and } \|H^{-1}g\| \leq \Delta, \\ -(H + \mu I)^{-1}g & \text{otherwise, where } \mu \geq \max(0, -\lambda_{\min}(H)) \text{ yields } \|p\| = \Delta. \end{cases}$$

6:   Compute reduction ratio:  $\rho \leftarrow \frac{f(\mathbf{x}) - f(\mathbf{x} + p)}{-(g^\top p + \frac{1}{2}p^\top H p)}$  ▷ Actual / predicted reduction
7:   if  $\rho > \eta$  then
8:      $\mathbf{x} \leftarrow \mathbf{x} + p$  ▷ Step accepted
9:   end if
10:  Radius Update:
11:  if  $\rho < 0.25$  then
12:     $\Delta \leftarrow 0.5\Delta$  ▷ Poor model agreement: Shrink radius
13:  else if  $\rho > 0.8$  and  $\|p\| = \Delta$  then
14:     $\Delta \leftarrow \min(2.5\Delta, \Delta_{\max})$  ▷ Excellent model agreement: Expand radius
15:  end if
16: until  $\|g\| \leq \epsilon_{\text{grad}}$  and  $\lambda_{\min}(H) \geq -\epsilon_{\text{nc}}$ 
17: return  $\mathbf{x}^* \leftarrow \mathbf{x}$ 

```

4.4. Zoom-In Penalty Continuation method (ZIPC) for Deep Null Synthesis

To balance the competing objectives of peak sidelobe minimization and null synthesis, we introduce a parameter continuation scheme on the null penalty weight, α . The augmented objective function is given by

$$f(\mathbf{x}; \alpha, \mu) = W(\mathbf{x}) + \alpha N(\mathbf{x}) + \mu C(\mathbf{x}),$$

where $W(\mathbf{x})$ is the sidelobe control functional defined in Section 3.6.1. We employ a continuation sequence that monotonically increases the penalty parameter, $\alpha \in \{1, 10^1, 10^2, \dots\}$, solving the Exact TR subproblem for each α_k while warm-starting from the preceding minimizer $\mathbf{x}_{k-1}^*$. The theoretical motivation for this approach relies on

the distinct local scaling behaviors of the functionals $W(\mathbf{x})$ and $N(\mathbf{x})$ near a baseline optimal design $\mathbf{x}^*$ (obtained at $\alpha = 1$). Because W is smooth, bounded perturbations within a sufficiently small neighborhood yield proportionally bounded variations in the peak sidelobe level. In contrast, the null functional $N(\mathbf{x})$ exhibits higher sensitivity within this same neighborhood. Because small perturbations in the element positions, and therefore phases on each scatterer, can fundamentally alter phase cancellation at specific angles, $N(\mathbf{x})$ can decrease by orders of magnitude even when the corresponding change in $W(\mathbf{x})$ is tightly bounded. The ZIPC method exploits this separation of sensitivities. By initially setting $\alpha = 1$, we drive the trajectory toward a feasible design characterized by strictly controlled sidelobes. As we increase α geometrically, the contribution of the null penalty dominates the gradient, forcing the optimizer along directions in the parameter space that dramatically reduce $N(\mathbf{x})$ while remaining within a tight level set of $W(\mathbf{x})$. In practice, this sequence is managed adaptively: we enforce a strict threshold T for the peak sidelobe functional (e.g., -28 dB). If a subproblem successfully converges while maintaining this performance bound, α is scaled up to further deepen the nulls. If the optimizer cannot find a feasible descent direction that respects the sidelobe constraint, the threshold is incrementally relaxed by a step size ΔT (e.g., 1 dB) to ensure continued progress. This method is full outlined in Algorithm 3. Note that ZIPC should be viewed as a continuation strategy for deepening prescribed nulls after a good sidelobe-controlled configuration has been found. It is not intended as a standalone global optimizer.

Algorithm 3 Zoom-In Penalty Continuation (ZIPC) for Null Synthesis

Require: Objective $f(\mathbf{x}; \alpha, \mu) = W(\mathbf{x}) + \alpha N(\mathbf{x}) + \mu C(\mathbf{x})$; Initial design $\mathbf{x}_0$;

Start penalty α_0 , growth factor $\gamma > 1$, max penalty $\alpha_{\max}$;

Peak-sidelobe target T , relaxation step $\Delta T > 0$

Ensure: Optimized design vector $\mathbf{x}^*$ balancing the objectives of deep nulls and low sidelobe levels.

```

1:  $\mathbf{x} \leftarrow \mathbf{x}_0$ 
2:  $\alpha \leftarrow \alpha_0$ 
3: repeat
4:   Update  $\mathbf{x}$  to a local minimum of  $f(\cdot; \alpha, \mu)$  via Algorithm 2, warm-started at  $\mathbf{x}$ 
5:   if  $\text{peakSidelobe}(\mathbf{x}) \leq T$  then
6:      $\alpha \leftarrow \gamma \alpha$  ▷ Success: Zoom in and deepen the null
7:   else
8:      $T \leftarrow T + \Delta T$  ▷ Target too aggressive: Relax threshold
9:   end if
10: until  $\alpha > \alpha_{\max}$ 
11: return  $\mathbf{x}^* \leftarrow \mathbf{x}$ 
    
```

5. Numerical Results

The existence results established that the constrained design problem is well-posed. However, the numerical experiments below should be interpreted as a study of the optimization landscape and the quality of attainable local minima rather than as a claim that global minimization has been solved algorithmically in all cases. The MultiStart computations provide an empirical benchmark for the best values found by extensive randomized search. The main numerical conclusion is that the physically informed concentric-ring initialization, together with second-order trust-region methods and ZIPC continuation, consistently reaches stable local minima whose objective values are close to the best MultiStart values at a fraction of the computational cost. All reported computations are within the fixed- z case with $2N$ positional variables. Table 2 summarizes the solver parameters used. We selected p_1, p_2 (see Section 3.7) through an L-curve analysis to approximate W and C as accurately as possible (by maximizing p_1 and p_2) while preventing numerical overflow.

5.1. Azimuthal Plane Optimization

We first restrict the optimization to control the radiation pattern solely in the azimuthal plane (i.e., radiation directions in the horizontal plane with the inclination angle fixed at $\theta = 90^\circ$). Controlling this 2D cut is computationally efficient and often yields a geometric configuration that behaves favorably in full 3D angular space.

Table 2
Local Solver Parameters.

ADAM Learning Rate α :	0.01
ADAM β_1, β_2 :	0.9, 0.99
trust-region initial radius Δ_0 :	1.0
trust-region acceptance threshold η :	0.1
LogSumExp parameter p_1 :	500
SoftPlus parameter p_2 :	300
penalty weights μ_1, μ_2 :	1, 100
ZIPC sequence for α :	$\{1, 10^1, 10^2, \dots\}$
stopping tolerances $\text{tol}_g, \text{tol}_{nc}$:	$10^{-7}, 10^{-8}$

5.1.1. Comparing Results for a Single Configuration

We evaluate the performance of all proposed optimizers on a representative test configuration. Note that we express angular coordinates in degrees throughout the following sections. The fixed parameters are: $N = 9$ elements, $\lambda_0 = 10$ m, $\Omega := \{90^\circ\} \times [-180^\circ, 180^\circ]$, with the main beam steered to $(\theta_0, \phi_0) = (90^\circ, 120^\circ)$, the source arriving from $(\theta_s, \phi_s) = (85^\circ, 120^\circ)$, and a null synthesized at $(\theta_{\text{null}}, \phi_{\text{null}}) = (90^\circ, 60^\circ)$. The main-beam exclusion radius M_ϕ is set to 60° and a mesh step size of 0.9° ($\pi/200$ radians) is used.

Radiation intensities are normalized to the peak power and converted to decibels via the formula:

$$U^{\text{dB}}(\mathbf{x}, (90^\circ, \phi)) = 10 \log_{10} \left(\frac{U(\mathbf{x}, (90^\circ, \phi))}{\max_{\phi' \in \Omega} U(\mathbf{x}, (90^\circ, \phi'))} \right). \quad (43)$$

Table 3 summarizes the convergence diagnostics, which are visualized in Fig. 10 in the Appendix. In these results, HPBW denotes the azimuthal half-power beamwidth, while CPUT and WCT represent CPU time and wall clock time, respectively. All computations were performed on a desktop computer equipped with an Intel Core Ultra 7 265 processor and 64 GB of RAM running Windows 11. Note that unlike the other optimizers, MultiStart was executed in parallel across 20 workers, averaging 45 minutes per run; meaning its serial equivalent is orders of magnitude slower than the local methods. Furthermore, the reported optimality criteria—gradient norm and minimum Hessian eigenvalue—carry different interpretations depending on whether the applied method optimizes the unconstrained penalized objective or the Lagrangian. Because the unconstrained methods minimize the smoothed aggregate spacing constraint C_{p_2} well below computational tolerance (and $C_{p_2} \geq C$), the strict spacing constraint, $C(\mathbf{x}) = 0$, is achieved by all solvers.

A notable outcome is that ADAM achieves nearly identical performance to the second-order fminunc solver, a result that remains consistent across the parameter sweep detailed in the following section. While ADAM is strictly a first-order method, its momentum-based updates allow it to effectively traverse the highly nonconvex landscape. However, in the full 3D optimization case discussed in Section 5.2, ADAM performs worse than the other methods, often stalling and exhibiting objective oscillation due to its fixed learning rate. This suggests that first-order adaptive methods can be competitive for lower-dimensional azimuthal cuts but become less reliable as the full 3D objective introduces additional curvature, saddle structure, and coupling between angular variables. Meanwhile, the ZIPC method achieves a substantially deeper null (below -270 dB) while maintaining peak sidelobe levels comparable to the other local methods, albeit at an increased computational cost.

Figs. 4 and 5 illustrate the Exact TR results. The polar plot demonstrates successful steering toward $\phi_0 = 120^\circ$ and a deep null at $\phi_{\text{null}} = 60^\circ$ with sidelobes in the azimuthal plane bounded below -30.27 dB. The full 3D pattern reveals that sidelobe levels outside the azimuthal plane were driven below -16.38 dB. However, because azimuthal only optimization leaves the elevation beamwidth unconstrained, the main beam is very broad in elevation. While this broadness highlights a key area for improvement in the full 3D pattern optimization it is partially expected; the array is physically small ($N = 9$) and relies on omnidirectional half-wave dipoles, which possess inherently low directivity. Consequently, an exclusion radius of $M_\phi = 60^\circ$ is appropriate, though results with $M_\phi = 45^\circ$ are viable in the case of azimuthal only optimization.

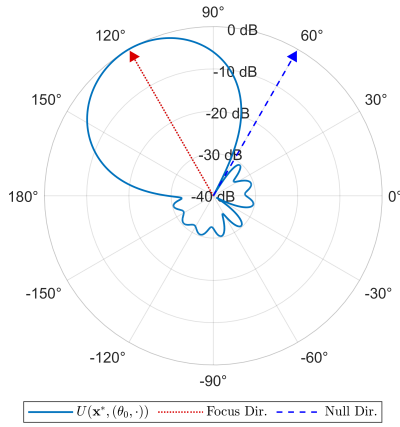
Figs. 6 and 7 display the MultiStart results. Although the azimuthal sidelobes are driven below -32.77 dB, the full 3D pattern exhibits the same broad elevation beamwidth seen in the Exact TR results and yields maximum 3D

sidelobe levels just below -14.3 dB. This trade-off is anticipated: as the optimizer converges on the best restricted 2D azimuthal objective, it naturally disregards out-of-plane 3D radiated energy.

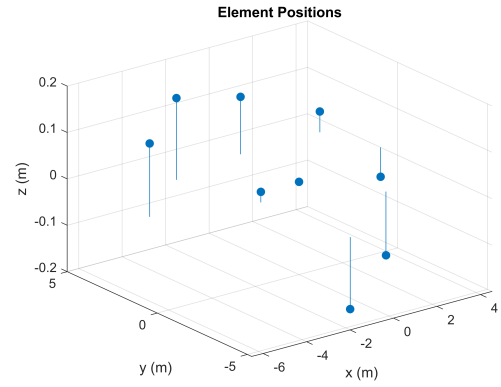
Table 3

Convergence diagnostics comparison between azimuthal optimizers on the fixed test configuration.

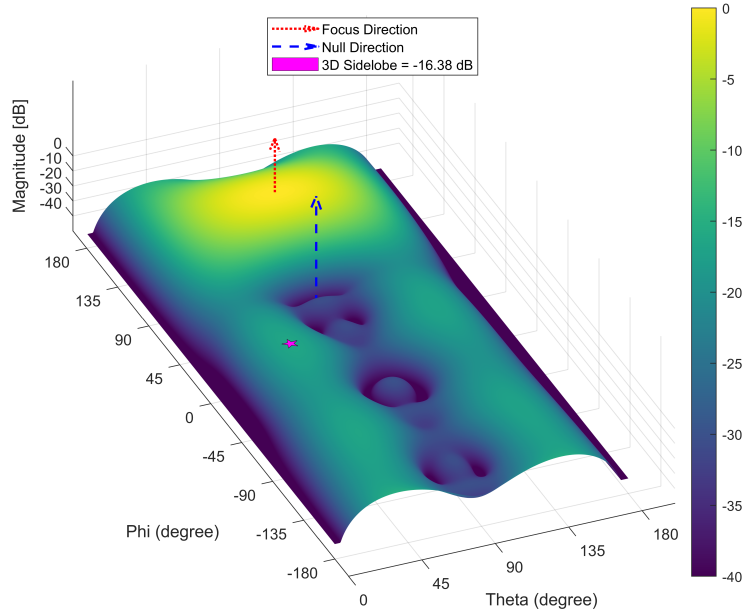
Optimizer	$W^{(2D)}(\mathbf{x}^*)$ [dB]	$N(\mathbf{x}^*)$ [dB]	$C(\mathbf{x}^*)$ [m]	$\ \nabla f\ _2$ [m^{-1}]	$\lambda_{\min}(\nabla^2 f)$ [m^{-2}]	HPBW [deg]	CPUT [s]	WCT [s]
ADAM	-28.44	-44.30	0	9.86×10^{-12}	-0.001923	48.636	16.500	2.5916
fmincon	-29.70	-40.00	0	8.29×10^{-12}	-0.008562	46.26	4.4063	0.6128
fminunc	-28.44	-44.30	0	6.55×10^{-6}	-0.001924	48.636	0.9531	0.1590
Exact TR	-30.27	-49.09	0	3.67×10^{-11}	0.003977	44.208	0.5625	0.1836
ZIPC	-30.14	-270.04	0	0.07657	0.003712	44.352	34.781	5.3047
MultiStart	-32.77	-50.42	0	4.95×10^{-6}	0.004335	49.428	9905.9	3429.4



(a) Azimuthal radiation pattern (polar plot).



(b) Optimized dipole element positions.

Fig. 4: Exact TR azimuthal optimization result.

Fig. 5: Full 3D radiation pattern (rectangular view) of the Exact TR azimuthal solution.

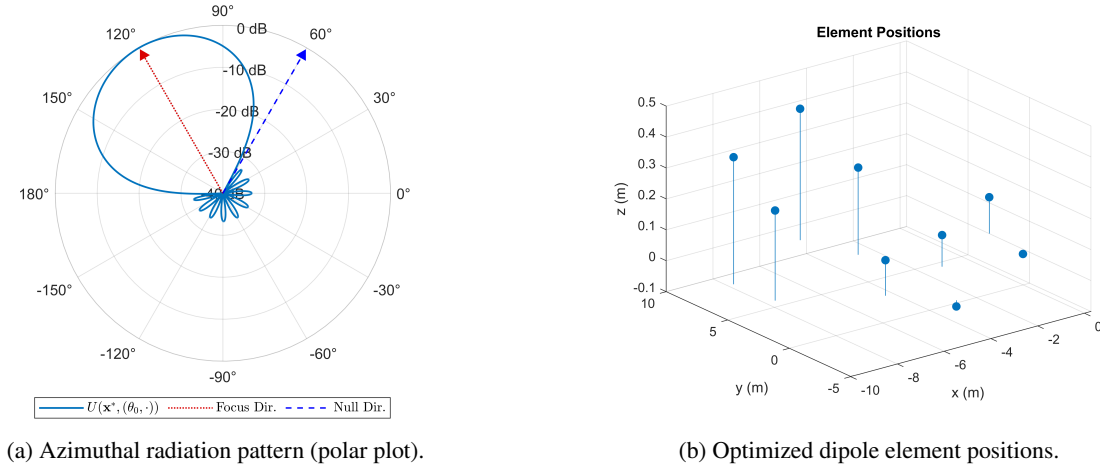


Fig. 6: MultiStart azimuthal optimization result.

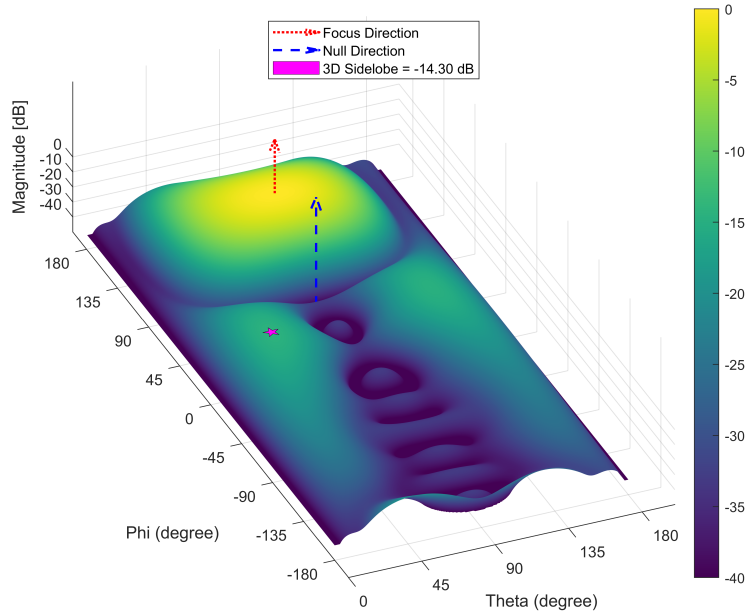


Fig. 7: Full 3D radiation pattern (rectangular view) of the MultiStart azimuthal solution.

5.1.2. Comparison of Algorithms across Parameter Sweep

To rigorously evaluate algorithmic robustness, we conducted a parameter sweep, varying the main beam focus direction ϕ_0 from 120° to 360° in 1° increments while synthesizing a null in a fixed direction $(\theta_{\text{null}}, \phi_{\text{null}}) = (90^\circ, 60^\circ)$. We keep $\phi_s = \phi_0$ throughout the sweep. Note that over this range the chosen null direction is always at least M_ϕ degrees away from the focus direction, so as not to destructively interfere with the main beam.

As shown in Fig. 8, performance remains stable across the full angular range. The methods are not merely tuned to a single favorable geometry, but generalize robustly. Interestingly, the sidelobe and null depths achieved by all optimizers exhibit periodic oscillations of similar phase. This rippling suggests an underlying harmonic structure in the objective landscape's geometry.

Across the sweep, `fmincon` achieved the lowest peak sidelobe levels for the majority of cases among the local methods. ZIPC reliably synthesized the deepest nulls while preserving comparable sidelobe levels. Average computation times over the sweep are compared in Table 4.

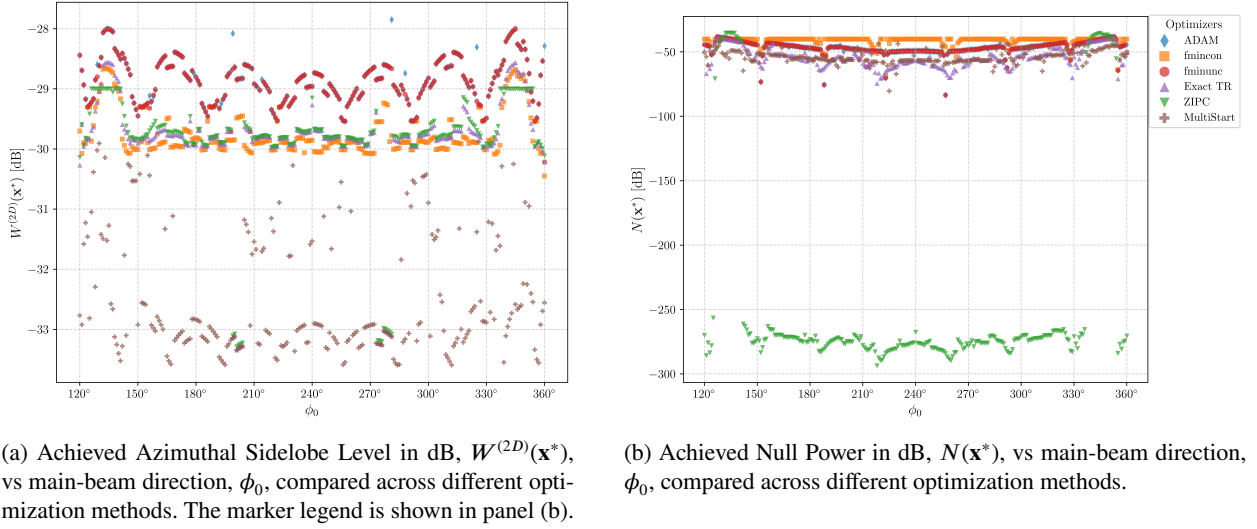


Fig. 8: Azimuthal optimization focus direction sweep results.

Table 4

Comparison of average computation time for each azimuthal optimizer over the sweep.

Optimizer	Avg. CPU Time [s]	Avg. Wall Clock Time [s]
ADAM	12.7676	1.9761
fmincon	4.2242	0.5937
fminunc	0.8509	0.1556
Exact TR	0.7748	0.1326
ZIPC	27.1679	4.3507
MultiStart	7509.4658	2774.9544

5.2. Full 3D Radiation Pattern Optimization

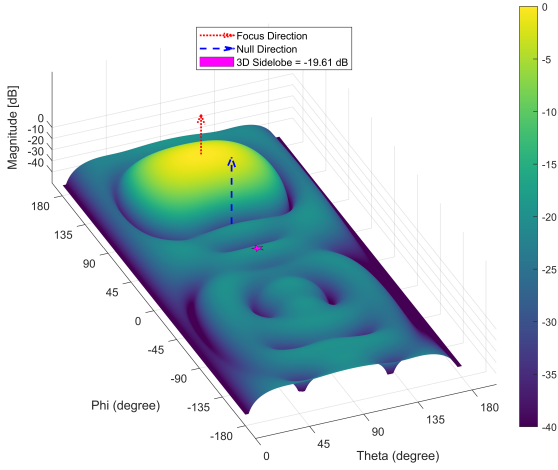
Finally, we optimize the complete 3D radiation pattern for the test configuration described in Section 5.1.1, but this time minimizing the peak sidelobe levels across all radiation directions $\Omega = [0, 180^\circ] \times [-180^\circ, 180^\circ]$ as opposed to an azimuthal pattern cut. We enforce main-beam exclusion radii of $M_\phi = M_\theta = 60^\circ$.

Convergence diagnostics for each optimizer are presented in Table 5 and Fig. 11. The reported timings demonstrate that extending the objective to 3D significantly increases the computational burden. However, as evidenced by the example ZIPC 3D optimization result in Fig. 9, controlling energy in the full spatial domain produces a main beam whose beamwidth is bounded in both the azimuthal and elevation planes. Notice that for all second-order methods, sidelobes are stably driven below -19 dB.

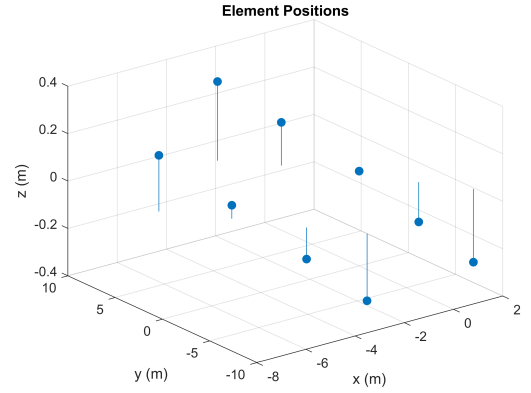
Table 5

Convergence diagnostics comparison between 3D optimizers on the fixed test configuration.

Optimizer	$W(\mathbf{x}^*)$ [dB]	$N(\mathbf{x}^*)$ [dB]	$C(\mathbf{x}^*)$ [m]	$\ \nabla f\ _2$ [m^{-1}]	$\lambda_{\min}(\nabla^2 f)$ [m^{-2}]	HPBW [deg]	CPUT [s]	WCT [s]
ADAM	-15.11	-29.82	0	1.26	-0.135493	43.34	344.3437	88.0606
fmincon	-19.62	-54.58	0	1.09×10^{-4}	0.098208	45.76	1687.1094	270.7531
fminunc	-19.57	-51.69	0	6.75×10^{-3}	0.186992	54.40	338.4375	86.5872
Exact TR	-19.62	-53.50	0	4.86×10^{-7}	0.384422	45.72	1091.0312	308.3600
ZIPC	-19.61	-267.42	0	0.1097	0.212293	45.58	3871.1250	974.3273



(a) 3D radiation pattern (rectangular view).



(b) Optimized dipole element positions.

Fig. 9: ZIPC 3D optimization result.

6. Conclusion

This paper studied the design of passive UAV relay arrays for directional beamforming and null synthesis through geometric optimization of relay locations. Unlike active distributed beamforming architectures, the proposed framework exploits the phase shifts induced by propagation path differences under known monochromatic illumination, so that coherent far-field shaping is achieved through spatial placement rather than oscillator synchronization. The design problem was formulated as a constrained nonlinear optimization problem in the three-dimensional positions of the passive relays, with objectives promoting energy concentration in a prescribed main beam and suppression in selected null directions, subject to geometric feasibility constraints such as minimum inter-relay separation. Within this framework, the beam pattern becomes a continuous function of the relay geometry, which allows the design task to be treated rigorously as a finite-dimensional variational problem. An existence result for an optimal configuration was established by combining compactness of the admissible set with continuity of the objective functional. Numerical investigations further showed that even relatively small relay swarms can achieve significant beam steering and null-forming capability when the relay positions are chosen appropriately. These computations illustrate the flexibility offered by mobile passive arrays and highlight the additional design freedom created by three-dimensional spatial reconfiguration. We emphasize the division of labor in these claims. The analysis shows that the problem admits a global minimizer, the proposed algorithms find stable, high-quality local minima, and MultiStart provides an empirical near-global benchmark that the structured concentric-ring initialization consistently approaches at a fraction of the computational cost. Overall, the results support the feasibility of passive UAV relay arrays as a mathematically structured and practically relevant architecture for adaptive far-field pattern control.

More broadly, these empirical observations provide practical guidance for geometry-driven scattering array design beyond the specific dipole array studied here. Because steering nulls and beams by relocating elements inherently creates an oscillatory, highly nonconvex optimization landscape, random initialization generally yields low success rates. We therefore recommend seeding the search with a symmetry-informed ansatz. Applying symmetric ring or lattice layouts alongside phase-matching conditions reliably targets a favorable basin prior to local refinement. Optimizing this ansatz over its symmetric parameters beforehand further improves overall efficiency. Additionally, rather than relying on the optimizer to discover relationships numerically, closed-form constraints—such as the beamfocusing condition used here—should be leveraged to explicitly eliminate decision variables. For configurations with small-to-moderate element counts, we recommend second-order trust-region solvers due to their demonstrated robustness against the saddles and plateaus defining these landscapes. First-order methods can be viable as computational shortcuts in reduced-angular-domain formulations, where variable coupling is weaker. Finally, when optimizing a smooth shaping objective alongside a highly sensitive cancellation objective (e.g., a deep null), our results indicate that a continuation strategy—progressively increasing the weight of the sensitive objective while warm-starting from intermediate solutions—provides an effective optimization framework.

A detailed sensitivity analysis is left for a subsequent study. In particular, the present work focuses on the formulation of the passive relay-array synthesis problem, the existence of optimal configurations, and the numerical construction of high-quality local minimizers. A systematic robustness investigation will require perturbing the optimized configurations with respect to UAV positioning errors, dipole-axis tilt errors, and manufacturing or deployment process errors, and then quantifying the resulting degradation in main-beam gain, peak sidelobe level, and null depth. Such an analysis is especially important for any practical implementation, but it is naturally separated from the present study because it involves a distinct uncertainty model and a separate set of Monte Carlo or worst-case simulations. We note, however, that the low-frequency regime considered here is favorable from this standpoint: for operating frequencies below 300 MHz, the free-space wavelength exceeds one meter, so perturbations on the order of a small fraction of a wavelength correspond to physically meaningful tolerances for UAV localization and orientation control.

A. Additional Figures

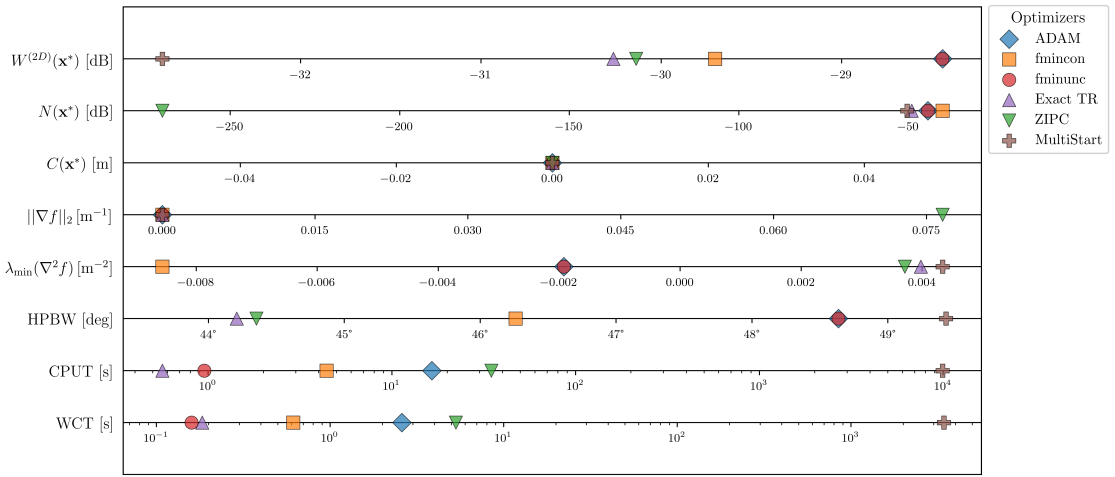


Fig. 10: Visual representation of the data presented in Table 3.

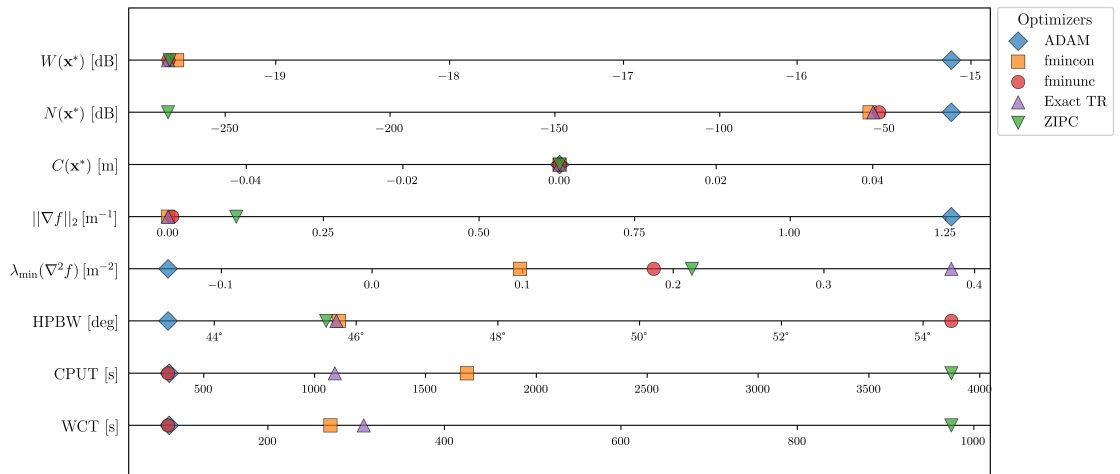


Fig. 11: Visual representation of the data presented in Table 5.

B. Berge Maximum Theorem

Here we review some basic definitions and conclude the section by citing the Berge Maximum Theorem in generality.

Definition 1 (Correspondence). A *correspondence* from a set X to a set Y is a map

$$\Gamma : X \rightrightarrows Y$$

which assigns to each element $x \in X$ a subset $\Gamma(x) \subseteq Y$. That is, $\Gamma(x) \subseteq Y$ for each $x \in X$.

Definition 2 (Upper Hemicontinuity). A correspondence $\Gamma : X \rightrightarrows Y$ is said to be *upper hemicontinuous* at $x_0 \in X$, if for every open set $V \subset Y$ such that $\Gamma(x_0) \subset V$, there exists a neighborhood U of x_0 such that,

$$\Gamma(x) \subset V \quad \text{for all } x \in U.$$

Equivalently, if $x_n \rightarrow x_0$ in X and $y_n \in \Gamma(x_n) \rightarrow y_0$ in Y , then $y_0 \in \Gamma(x_0)$.

Definition 3 (Lower Hemicontinuity). A correspondence $\Gamma : X \rightrightarrows Y$ is *lower hemicontinuous* at $x_0 \in X$, if for every open set $V \subset Y$ such that $\Gamma(x_0) \cap V \neq \emptyset$, there exists a neighborhood U of x_0 such that,

$$\Gamma(x) \cap V \neq \emptyset \quad \text{for all } x \in U.$$

Equivalently, if $x_n \rightarrow x_0$ in X and $y_0 \in \Gamma(x_0)$, then there exists $y_n \in \Gamma(x_n)$ such that $y_n \rightarrow y_0$ in Y .

Theorem 7 (Berge's Maximum Theorem [2, 5]). Let X and Y be topological spaces. Let $F : X \times Y \rightarrow \mathbb{R}$ be a continuous function, and suppose $\Gamma : X \rightrightarrows Y$ is a correspondence such that for each $x \in X$ the set $\Gamma(x) \subset Y$ is nonempty and compact. Define the value function

$$f(x) = \max_{y \in \Gamma(x)} F(x, y),$$

and the argmax correspondence $M : X \rightrightarrows Y$,

$$M(x) := \arg \max_{y \in \Gamma(x)} F(x, y).$$

Then:

- (i) If Γ is upper hemicontinuous, then $f : X \rightarrow \mathbb{R}$ is upper semicontinuous and M is nonempty, compact-valued, and upper hemicontinuous.
- (ii) If Γ is lower hemicontinuous, then $f : X \rightarrow \mathbb{R}$ is lower semicontinuous.
- (iii) Consequently, if Γ is continuous (that is, both upper and lower hemicontinuous), then f is continuous and M is nonempty, compact-valued, and upper hemicontinuous.

Part (iii) is the form of the theorem invoked in Theorem 4: continuity of PS requires both hemicontinuity properties of Γ , which is why Lemma 3 establishes lower hemicontinuity separately on $X_0(\xi, \kappa)$.

C. Gradient and Hessian of f_μ

C.1. Notation and Basic Identities

Let $R = (\mathbf{r}_1, \dots, \mathbf{r}_N)^\top \in \mathbb{R}^{N \times 3}$ where $\mathbf{r}_m = (x_m, y_m, z_m)^\top$. For brevity, we define the wave vector difference:

$$\mathbf{C}(\theta, \phi) = \mathbf{k}(\theta, \phi) - \mathbf{k}(\theta_s, \phi_s). \quad (44)$$

Furthermore, let:

$$c_m := \cos(\mathbf{C} \cdot \mathbf{r}_m), \quad s_m := \sin(\mathbf{C} \cdot \mathbf{r}_m), \quad (45)$$

and their corresponding sums across all elements:

$$\alpha := \sum_{n=1}^N c_n, \quad \beta := \sum_{n=1}^N s_n. \quad (46)$$

We define the vector $\boldsymbol{\gamma} \in \mathbb{R}^N$ such that its m -th component is given by:

$$\gamma_m := \beta c_m - \alpha s_m. \quad (47)$$

The radiation power density at direction (θ, ϕ) is then:

$$U(\mathbf{x}, (\theta, \phi)) = e(\theta)^2 \left| \sum_{n=1}^N \exp(-i\mathbf{C} \cdot \mathbf{r}_n) \right|^2 = e(\theta)^2 (\alpha^2 + \beta^2). \quad (48)$$

C.2. Gradients of U

C.2.1. Free-z Case

In this case z does not depend on x or y . A direct computation gives the partial derivatives componentwise. For $w \in \{x, y, z\}$ we have,

$$\partial_{w_m} U = 2C_w e(\theta)^2 (\beta c_m - \alpha s_m) = 2C_w e(\theta)^2 \gamma_m, \quad (49)$$

where C_w denotes the w component of $\mathbf{C}$. Stacking these partial derivatives over $m = 1, \dots, N$ yields the gradient in matrix form:

$$\nabla_R U = 2e(\theta)^2 \boldsymbol{\gamma} \mathbf{C}^\top, \quad (50)$$

where the (m, w) -th entry is precisely $(\nabla_R U)_{m,w} = 2e(\theta)^2 C_w \gamma_m$. Note that $\text{rank}(\nabla_R U) \leq 1$.

C.2.2. Fixed-z Case

When z is determined by the beamfocusing condition, z is linearly dependent on x_n and y_n . Thus, define

$$D_x := \partial_{x_n} z_n, \quad D_y := \partial_{y_n} z_n \quad (\text{independent of } n). \quad (51)$$

We define the effective steering vector as

$$\hat{\mathbf{C}} := (C_x + C_z D_x, C_y + C_z D_y) \in \mathbb{R}^2. \quad (52)$$

Then, for $w \in \{x, y\}$,

$$\partial_{w_m} U = 2(C_w + C_z D_w) e(\theta)^2 (\beta c_m - \alpha s_m). \quad (53)$$

This yields the fixed-z gradient:

$$\nabla_R U = 2e(\theta)^2 \boldsymbol{\gamma} \hat{\mathbf{C}}^\top. \quad (54)$$

C.3. Hessians of U

C.3.1. Free-z Case

Let δ_{ij} be the Kronecker delta. Then, for $v, w \in \{x, y, z\}$,

$$\partial_{v_b} \partial_{w_m} U = 2 C_w C_v e(\theta)^2 \sum_{n=1}^N \cos(\mathbf{C} \cdot (\mathbf{r}_n - \mathbf{r}_m)) (\delta_{nb} - \delta_{mb}). \quad (55)$$

Hence $H(\mathbf{x}, (\theta, \phi))$, the Hessian of U taken with respect to the coordinate-major ordering of the position variables (i.e. the column-wise vectorization of $\mathbf{R}$, in which all x -coordinates precede all y -coordinates, which in turn precede all z -coordinates), has the Kronecker structure

$$H(\mathbf{x}, (\theta, \phi)) = 2 e(\theta)^2 (\mathbf{C} \mathbf{C}^\top) \otimes M(\mathbf{x}, (\theta, \phi)) \quad (56)$$

where $M \in \mathbb{R}^{N \times N}$ is defined entrywise as

$$M_{mb} = \begin{cases} \cos(\mathbf{C} \cdot (\mathbf{r}_b - \mathbf{r}_m)), & b \neq m, \\ 1 - \sum_{n=1}^N \cos(\mathbf{C} \cdot (\mathbf{r}_n - \mathbf{r}_m)), & b = m. \end{cases} \quad (57)$$

Letting $\mathbf{c} = (c_1, \dots, c_N)^\top$ and $\mathbf{s} = (s_1, \dots, s_N)^\top$, we can express M as

$$M = \mathbf{c}\mathbf{c}^\top + \mathbf{s}\mathbf{s}^\top - \text{diag}(\alpha\mathbf{c} + \beta\mathbf{s}). \quad (58)$$

The Kronecker product structure of H allows for efficient matrix assembly and code vectorization. Note also that $\text{rank}(M) \leq 2$.

C.3.2. Fixed-z Case

The Hessian of U (w.r.t the position variables) retains the same Kronecker structure as above simply replacing $\mathbf{C}$ with $\hat{\mathbf{C}}$:

$$H(\mathbf{x}, (\theta, \phi)) = 2e(\theta)^2 (\hat{\mathbf{C}}\hat{\mathbf{C}}^\top) \otimes M(\mathbf{x}, (\theta, \phi)). \quad (59)$$

C.4. Derivatives of W_{p_1}

Recall the definition of the smoothed W functional:

$$W_{p_1}(x) = \frac{1}{p_1} \log \int_I \exp(p_1 U(x, \omega)) d\omega. \quad (60)$$

Then, by the Leibniz integral rule,

$$\nabla W_{p_1}(x) = \frac{\int_I \exp(p_1 U(x, \omega)) \nabla U(x, \omega) d\omega}{\int_I \exp(p_1 U(x, \omega)) d\omega}. \quad (61)$$

For the Hessian we have

$$\nabla^2 W_{p_1}(x) = \frac{\int_I \exp(p_1 U) \nabla^2 U d\omega}{\int_I \exp(p_1 U) d\omega} + p_1 \left[\frac{\int_I \exp(p_1 U) \nabla U \nabla U^\top d\omega}{\int_I \exp(p_1 U) d\omega} - \nabla W_{p_1}(x) \nabla W_{p_1}(x)^\top \right]. \quad (62)$$

C.5. Derivatives of the Smoothed Aggregate Spacing Constraint C_{p_2}

Let $d_{n,m} = \|\mathbf{r}_n - \mathbf{r}_m\|_2$ denote the pairwise distance. Then,

$$C_{n,m} = \frac{\lambda_0}{4} - d_{n,m}, \quad C_{p_2}(\mathbf{x}) = \frac{1}{p_2} \sum_{n < m} \log(1 + \exp(p_2 C_{n,m})) = \sum_{n < m} \text{SoftPlus}_{p_2}(C_{n,m}). \quad (63)$$

Using the standard sigmoid function $\sigma(t) := (1 + \exp(-t))^{-1}$ and its derivative $\sigma'(t) = \sigma(t)(1 - \sigma(t))$, we have that

$$\partial_{v_k} C_{p_2}(\mathbf{x}) = \sum_{n < m} \sigma(p_2 C_{n,m}) \partial_{v_k} C_{n,m}. \quad (64)$$

The corresponding mixed partial derivatives for the Hessian are:

$$\partial_{w_j} \partial_{v_k} C_{p_2} = \sum_{n < m} \left[p_2 \sigma(p_2 C_{n,m}) (1 - \sigma(p_2 C_{n,m})) (\partial_{w_j} C_{n,m}) (\partial_{v_k} C_{n,m}) + \sigma(p_2 C_{n,m}) \partial_{w_j} \partial_{v_k} C_{n,m} \right]. \quad (65)$$

Since $C_{n,m} = \lambda_0/4 - d_{n,m}$, it follows immediately that $\partial C_{n,m} = -\partial d_{n,m}$ and $\partial^2 C_{n,m} = -\partial^2 d_{n,m}$.

C.5.1. Derivatives of $d_{n,m}$ (Free- z Case)

We have $d_{n,m} = \sqrt{(x_n - x_m)^2 + (y_n - y_m)^2 + (z_n - z_m)^2}$. For $n \neq m$ the first derivatives are

$$\frac{\partial d_{n,m}}{\partial w_j} = \frac{(\delta_{nj} - \delta_{mj})(w_n - w_m)}{d_{n,m}}, \quad (66)$$

and the second derivatives are

$$\frac{\partial^2 d_{n,m}}{\partial v_k \partial w_j} = \frac{(\delta_{nj} - \delta_{mj})(\delta_{nk} - \delta_{mk})}{d_{n,m}^3} \left[d_{n,m}^2 \delta_{vw} - (w_n - w_m)(v_n - v_m) \right]. \quad (67)$$

C.5.2. Derivatives of $d_{n,m}$ (Fixed- z Case)

As before z_n depends linearly on (x_n, y_n) with constants $D_x = \partial_{x_n} z_n$, $D_y = \partial_{y_n} z_n$. For $w \in \{x, y\}$ define

$$\Delta w := w_n - w_m, \quad \Delta z := z_n - z_m, \quad \widetilde{\Delta w} := \Delta w + D_w \Delta z, \quad a_j := \delta_{nj} - \delta_{mj}. \quad (68)$$

Then, for $n \neq m$,

$$\frac{\partial d_{n,m}}{\partial w_j} = \frac{a_j \widetilde{\Delta w}}{d_{n,m}}, \quad \frac{\partial^2 d_{n,m}}{\partial v_k \partial w_j} = \frac{a_j a_k}{d_{n,m}^3} \left[(\delta_{vw} + D_w D_v) d_{n,m}^2 - \widetilde{\Delta w} \widetilde{\Delta v} \right]. \quad (69)$$

CRedit authorship contribution statement

Dustin Bryant: Writing – review & editing, Writing – original draft, Software, Literature Review, Investigation. **Daniil Shaposhnikov:** Writing – original draft, Writing – review & editing, Visualization, Validation, Software, Investigation. **Malcolm Davies:** Writing – review & editing, Writing – original draft, Software, Literature Review, Investigation. **Daniel Onofrei:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization. **David R. Jackson:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.

During the preparation of this work the authors used Claude Code (model Opus 4.5) in order to assist in the literature review and write portions of the introduction. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The source code and optimization result datasets from this study are available on Zenodo [38].

References

- [1] Abdelhay, M.A., 2025. A convex optimization-based approach for sidelobe level suppression and null control in antenna arrays by displacing a minimum number of elements. *Journal of Telecommunications and Information Technology* 101, 63–68. URL: <https://jtit.pl/jtit/article/view/2198>, doi:10.26636/jtit.2025.3.2198.
- [2] Aliprantis, C.D., Border, K.C., 2006. *Infinite Dimensional Analysis: A Hitchhiker's Guide*. Springer, Berlin; London. doi:10.1007/3-540-29587-9.
- [3] Alsarayreh, S., Mohamed Radzi, N.A., Alkahtani, A., Mohd Azmi, K.H., 2023. Intelligent reflecting surface-UAV systems in the internet of things network: A survey. *IEEE Access* PP, 1–1. URL: https://www.researchgate.net/publication/372919623_Intelligent_Reflecting_Surface-UAV_Systems_in_the_Internet_of_Things_Network_A_Survey, doi:10.1109/ACCESS.2023.3301959.
- [4] Balanis, C.A., 2016. *Antenna Theory: Analysis and Design*. 4 ed., Wiley, Hoboken, NJ. URL: <https://www.wiley.com/en-us/Antenna+Theory%3A+Analysis+and+Design%2C+4th+Edition-p-9781118642061>.
- [5] Berge, C., 1963. *Topological Spaces: Including a Treatment of Multi-valued Functions, Vector Spaces, and Convexity*. Oliver & Boyd, Edinburgh and London.
- [6] Conn, A.R., Gould, N.I.M., Toint, P.L., 2000. *Trust-Region Methods*. MPS-SIAM Series on Optimization, SIAM, Philadelphia, PA.
- [7] Cui, M., Cai, Y., Wu, Q., Chen, W., Xu, J., Zhang, R., Ng, D.W.K., 2021. Joint trajectory and passive beamforming design for IRS-assisted UAV communications. *IEEE Journal on Selected Areas in Communications* 39, 3072–3089. doi:10.1109/JSAC.2021.3088623.
- [8] Cui, M., Cai, Y., Xu, J., Zhang, R., Ng, D.W.K., 2024. IRS-assisted UAV communications: A comprehensive survey. *IEEE Communications Surveys & Tutorials* 26, 1086–1124. doi:10.1109/COMST.2024.3353300.
- [9] Dabiri, M.T., Hasna, M.O., Khattab, T., Qaraqe, K., 2022. A study of multihop mmW aerial backhaul links, in: *2022 International Wireless Communications and Mobile Computing (IWCMC)*, pp. 1228–1233. URL: <https://arxiv.org/pdf/2203.04387>, doi:10.1109/IWCMC55113.2022.9824118.
- [10] Diao, J., Hedayati, M., Wang, H., Huang, X., Sunar, B., Holzman, J.F., 2019. Experimental demonstration of distributed beamforming on two flying mini-drones, in: *2019 USNC-URSI Radio Science Meeting (Joint with IEEE AP-S Symposium)*, Atlanta, GA, USA. pp. 117–118. URL: <https://genesys-lab.org/papers/1570588434.pdf>, doi:10.23919/USNC-URSI.2019.8861933.
- [11] Dolph, C.L., 1946. A current distribution for broadside arrays which optimizes the relationship between beam width and side-lobe level. *Proceedings of the IRE* 34, 335–348. URL: <https://ieeexplore.ieee.org/document/1697083>, doi:10.1109/JRPROC.1946.225956.
- [12] Egarguin, N.J.A., Jackson, D.R., Onofrei, D., Leclerc, J., Becker, A., 2020. Adaptive beamforming using scattering from a drone swarm, in: *2020 IEEE Texas Symposium on Wireless and Microwave Circuits and Systems (WMCS)*, Waco, TX, USA. pp. 1–6. URL: <https://par.nsf.gov/servlets/purl/10219784>, doi:10.1109/WMC549442.2020.9172335.
- [13] Haupt, R.L., 1994. Thinned arrays using genetic algorithms. *IEEE Transactions on Antennas and Propagation* 42, 993–999. URL: <https://ui.adsabs.harvard.edu/abs/1994ITAP...42..993H/abstract>, doi:10.1109/8.299611.
- [14] Huang, J., Encinar, J.A., 2007. *Reflectarray Antennas*. Wiley-IEEE Press, Hoboken, NJ. URL: <https://www.wiley.com/en-us/Reflectarray+Antennas-p-9780470084915>.
- [15] Imaz-Lueje, B., Prado, D.R., Arrebola, M., Pino, M.R., 2020. Reflectarray antennas: a smart solution for new generation satellite mega-constellations in space communications. *Nature, Scientific Reports* URL: <https://www.nature.com/articles/s41598-020-78501-0>, doi:10.1038/s41598-020-78501-0.
- [16] Jackson, J.D., 1999. *Classical Electrodynamics*. 3 ed., Wiley, New York. URL: <https://ieee.li/pdf/essay/classical-electrodynamics-john-david-jackson.pdf>.
- [17] Liu, C., Mei, W., Wang, P., Meng, Y., Ning, B., Chen, Z., 2024. UAV-enabled passive 6D movable antennas: Joint deployment and beamforming optimization. *arXiv:2412.11150*. arXiv:2412.11150.
- [18] Ma, W., Zhu, L., Zhang, R., 2023. Multi-beam forming with movable-antenna array. *arXiv:2311.03775*. arXiv:2311.03775.
- [19] Mailloux, R.J., 2017. *Phased Array Antenna Handbook*. 3 ed., Artech House.
- [20] Merlo, J.M., Mghabghab, S.R., Nanzer, J.A., 2023. Wireless picosecond time synchronization for distributed antenna arrays. *IEEE Transactions on Microwave Theory and Techniques* 71, 1720–1731. URL: <https://ieeexplore.ieee.org/document/9994246>, doi:10.1109/TMTT.2022.3227878.
- [21] Mghabghab, S.R., Nanzer, J.A., 2021. Open-loop distributed beamforming using wireless frequency synchronization. *IEEE Transactions on Microwave Theory and Techniques* 69, 896–905. URL: <https://ieeexplore.ieee.org/document/9210120>, doi:10.1109/TMTT.2020.3022385.
- [22] Mohanti, S., Bocanegra, C., Meyer, J., Secinti, G., Diddi, M., Singh, H., Chowdhury, K., 2019. Airbeam: Experimental demonstration of distributed beamforming by a swarm of uavs, in: *2019 IEEE 16th International Conference on Mobile Ad Hoc and Sensor Systems (MASS)*, IEEE. pp. 162–170. doi:10.1109/MASS.2019.00028.
- [23] Moré, J.J., Sorensen, D.C., 1983. Computing a trust region step. *SIAM Journal on Scientific and Statistical Computing* 4, 553–572. doi:10.1137/0904038.
- [24] Mudumbai, R., Barriac, G., Madhow, U., Hespanha, J.P., 2010. Distributed transmit beamforming using feedback control. *IEEE Transactions on Information Theory* 56, 411–426. URL: <https://ieeexplore.ieee.org/document/5361473>, doi:10.1109/TIT.2009.2034786.
- [25] Mudumbai, R., Hespanha, J., Madhow, U., Barriac, G., 2005. Scalable feedback control for distributed beamforming in sensor networks, in: *Proc. IEEE Int. Symp. on Information Theory (ISIT)*, Adelaide, SA, Australia. pp. 137–141. URL: https://wsl.ece.ucsb.edu/sites/default/files/publications/scalablefeedback_raghu.pdf, doi:10.1109/ISIT.2005.1523308.
- [26] Nanzer, J., Mghabghab, S., Ellison, S., Schlegel, A., 2021. Distributed phased arrays: Challenges and recent advances. *IEEE Transactions on Microwave Theory and Techniques* 69, 4893–4907. URL: <https://api.semanticscholar.org/CorpusID:237985334>, doi:10.1109/TMTT.2021.3092401.

- [27] Ohira, T., 2002. Analog smart antennas: An overview, in: Proceedings of the 13th IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), Lisbon, Portugal. pp. 1502–1506.
- [28] Ohira, T., Gyoda, K., 2000. Electronically steerable passive array radiator antennas for low-cost analog adaptive beamforming, in: Proceedings of the 2000 IEEE International Conference on Phased Array Systems and Technology, Dana Point, CA, USA. pp. 101–104. doi:10.1109/PAST.2000.858918.
- [29] Ouassal, A., Yan, Z., Nanzer, J.A., 2021. Decentralized frequency alignment for collaborative beamforming in distributed phased arrays. *IEEE Transactions on Wireless Communications* 20, 6269–6281. URL: <https://arxiv.org/html/1911.10076>, doi:10.1109/TWC.2021.3076805.
- [30] Park, K.W., Kim, H.M., Shin, O.S., 2022. A survey on intelligent-reflecting-surface-assisted UAV communications. *Energies* 15, 5143. URL: <https://doi.org/10.3390/en15145143>, doi:10.3390/en15145143. open access under CC BY 4.0 license.
- [31] Petko, J.S., Werner, D.H., 2009. Positional tolerance analysis and error correction of micro-UAV swarm based antenna arrays, in: Proc. IEEE Antennas and Propagation Society Int. Symp. (APS), Charleston, SC, USA. pp. 2547–2550. URL: <https://www.researchgate.net/publication/224566540>. earlier versions also listed with 2008 date in some bibliographies.
- [32] Rahmatov, N., Baek, M., 2023. RIS-carried UAV communication: Current research, challenges, and future opportunities. *ICT Express* 9, 556–569. URL: <https://doi.org/10.1016/j.ictex.2023.04.005>, doi:10.1016/j.ictex.2023.04.005.
- [33] Rashid, M., Nanzer, J.A., 2023. Frequency and phase synchronization in distributed phased arrays. *IEEE Transactions on Wireless Communications* 22, 364–378. URL: <https://ieeexplore.ieee.org/document/9923618>, doi:10.1109/TWC.2022.3213788.
- [34] Robinson, J., Rahmat-Samii, Y., 2004. Particle swarm optimization in electromagnetics. *IEEE Transactions on Antennas and Propagation* 52, 397–407. URL: <https://ieeexplore.ieee.org/document/1282114>, doi:10.1109/TAP.2004.823969.
- [35] Schelkunoff, S.A., 1943. A mathematical theory of linear arrays. *Bell System Technical Journal* 22, 80–107. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/j.1538-7305.1943.tb01306.x>, doi:10.1002/j.1538-7305.1943.tb01306.x.
- [36] Schlegel, A., Merlo, J.M., Nanzer, J.A., 2023. Coherent distributed bistatic radar using wireless frequency syntonization and internode ranging. *IEEE Microwave and Wireless Technology Letters* 33, 1393–1396. URL: <https://ieeexplore.ieee.org/document/10185098>, doi:10.1109/LMWT.2023.3283918.
- [37] Shang, B., Shafin, R., Liu, L., 2021. UAV swarm-enabled aerial reconfigurable intelligent surface. *arXiv preprint arXiv:2103.06361*. URL: <https://arxiv.org/abs/2103.06361>. submitted 10 Mar 2021.
- [38] Shaposhnikov, D., Bryant, D., Davies, M., 2026. Passive relay array geometry optimization. URL: <https://doi.org/10.5281/zenodo.21800059>, doi:10.5281/zenodo.21800059. zenodo dataset, version 1.0.
- [39] Stratton, J.A., 1941. *Electromagnetic Theory*. McGraw-Hill, New York. URL: <https://ieeexplore.ieee.org/book/7362911>.
- [40] Taylor, T.T., 1955. Design of line-source antennas for narrow beamwidth and low side-lobe levels. *IRE Transactions on Antennas and Propagation* AP-3, 16–28. URL: <https://www.scirp.org/reference/referencespapers?referenceid=1477682>, doi:10.1109/TPGAP.1955.5720407.
- [41] Van Trees, H.L., 2002. *Optimum Array Processing: Part IV of Detection, Estimation, and Modulation Theory*. Wiley-Interscience, New York. URL: https://www.researchgate.net/publication/235267581_Optimum_Array_Processing_-_Part_IV_of_Detection_Estimation_and_Modulation_Theory, doi:10.1002/0471221104.
- [42] Zarifi, K., Affes, S., Ghayeb, A., 2010. Collaborative null-steering beamforming for uniformly distributed wireless sensor networks. *IEEE Transactions on Signal Processing* 58, 1889–1903. URL: https://www.wirelesslab.ca/File/pdf_files/journals/J046.pdf#:~:text=Collaborative%20Null,with%20the%20decen%02tralized%20nature%20of, doi:10.1109/TSP.2009.2036476.
- [43] Zhu, L., Ma, W., Ning, B., Zhang, R., 2023a. Movable-antenna enhanced multiuser communication via antenna position optimization. *arXiv:2302.06978*. arXiv:2302.06978.
- [44] Zhu, L., Ma, W., Zhang, R., 2023b. Movable-antenna array enhanced beamforming: Achieving full array gain with null steering. *IEEE Communications Letters* 27, 3340–3344. doi:10.1109/LCOMM.2023.3323656.